

2nd Applied Active Inference Symposium on "Robotics" ~ 1st session

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[Music] hello and welcome everyone to the second applied active inference symposium hosted by the active inference institute it is july 31st 2022 and this is the first session of the symposium the focus of the symposium will be robotics and the presentations will be centered around that theme if you have ideas for future symposium topics and want to participate in organizing please reach out to us for those of you watching live please post questions in the chat and we will ask the presenters during the roundtable discussion this symposium will be recorded transcribed and archived for lasting access we will make the playlist available for asynchronous participation if you would like to participate in the transcription of the video please reach out to us at activeinference@gmail.com we will have five presenters followed by a roundtable discussion the presenters in the first block are going to be tim schneider presenting active inference for robotic manipulation with co-authors let's see sorry oh i don't have any listed here um okay and then next let's see next is tim verbellen um and he is presenting robotics modeling the world from pixels using deep active inference also no co-authors listed and then next will be next will be ben white with artificial empathy active inference and collective intelligence and he has co-authors mark miller and daphne damascus demacos sorry if i got that wrong and after that we will have a talk by noor sajid learning agent preferences let's see her are ah there she is oh she doesn't have any listed here either and then finally we have one hua chen with the talk called dual control for exploitation and exploration and its applications in robotic autonomous search and that's it so with pleasure i introduce tim schneider uh please take it away yeah thanks a lot for the introduction i'm just quickly going to share my screen i hope you can see that all right yeah so my name is tim schneider and today i want to talk about our work on active influence for robotic manipulation so i think we can all agree that so there's some noise on the background sorry yeah it never went i think we can all agree that uh manipulation is one of these central abilities that we need in our everyday life like be it cooking writing or using tools and you can obviously think of a variety of other tasks

that also require dexterous however despite this significance of manipulation in our everyday lives robotic manipulation is still a largely unsolved topic i think one of the main reasons for this is that usually in classic robotics how we did it for years we always assume that everything is kind of known that we know where everything is how everything behaves but in unstructured environments this is usually not the case and so what we need is very adaptive policies that are able to react to changes in the environment and are robust to to all kind of perturbations so what my lab focuses on or at least in part focuses on is applying reinforcement learning to robotic manipulation so learn these skills instead of programming them by hand however this is also not super straightforward in manipulation and one of the central challenges here is to to perform the exploration so in reinforcement learning we always have to explore a task before we can complete it like for example here in this task um this robot has to move up this little ball into a target zone on this tilted table and what we usually do and what is done here is that we just apply some random actions in the beginning and also throughout the the entire optimization procedure and we just hope that this will give us some useful insight into how the world actually behaves and how we can create a high reward in these settings but i think in manipulation this is usually not the case because if we just apply random noise we end up dropping the objects we are trying to manipulate we end up maybe destroying even parts of the environment so i think this approach is simply not feasible for interacting with the real world in and if we look at how humans explore the world we see a very different picture um so this is a toddler that has obviously the task of building the highest possible tower of these other piece blocks and what we can see is that this toddler is not simply applying some random noisy actions this kid has actually a fairly directed way of exploring it has an idea of what is useful to learn what might bring it forward or maybe maybe this is not as planned but more of an intuition but but certainly this exploration is much smarter than what we are currently doing in reinforcement learning and so to summarize this exploration is a very challenging task in reinforcement learning for humans if we take them as an example explore very actively and in a very directed fashion and the question that we want to ask in this in this work is can we do this with robots too and so this is where we started looking into cognitive science and also came across active inference mainly because it uh it proposes a way well it proposes a theory of explaining curiosity in intelligent beings and the question was whether we can translate this on onto robots so i just for completeness i want to quickly go over the basics so in active inference we assume that every agent maintains some kind of model of the world which consists of observations o hidden states x that the agent

cannot observe and some actions or policy π and the objective is to minimize the price which is the negative log likelihood of the marginal observation probability under the model of the agent and it has the agent has two ways of doing this the first one is inference so understanding the causes of the observations it is making this is done by applying variation to compute an upper bound of this objective which is called variational free energy and then we minimize this which corresponds to finding a variational posterior for hidden states given the observations we are currently making and the other avenue which is what got us interested in the topic is um the selection of actions in order to minimize this free energy objective uh so if we want to do this we have to make sure that we kind of plan ahead so we have to take an expectation of future states we might encounter given that we choose some form of action and we take this expectation of this free energy term and what we get out is fairly interesting namely we get this kind of formulation here which decomposes into an expected information game term and an extrinsic term and the way this expected information gain term works or what it encourages is to take informative actions because this term becomes maximal when we take an action which causes us to learn as much as possible about the current latent state of the world so for example if you're in a very dark room a very informative action would be to turn the light on it's not immediately goal directed or anything but it will at least tell you where you are what your surroundings look like and and so on and then the second term is this extrinsic term and here the agent maintains a preference distribution over observations this is a bit of an unorthodox thing for me as a reinforcement learning person we usually simply define a reward but here this extrinsic preference is defined in terms of of this target observation distribution but this is also exactly the point where we can inject a target behavior into the agent by basically saying okay this agent should prefer to observe some specific observation now taking a step back and going to the reinforcement learning part of this talk again so the problem statement that we are looking at is a finite horizon mdp which means that there are states actions and rewards we assume that these states fully encode the entire state of the environment so this is there's no hidden state in the environment and the objective of the agent is to maximize its expected return for one episode and it's the challenge here is that the dynamics and the reward distribution so these two distributions here are completely unknown um so the only way the agent has to figure those out is to probe the environment with different observe the outcomes and then learn some kind of model out of this i mean obviously there are also other ways of doing this this model doesn't have to be learned but in our case it will be learned now we can see reinforcement learning through the lens of active inference i just

wrote down this mvp again from the previous slide here the first thing we need is to define the internal model of the agent and we do this in a way that we say okay there's two models the agent maintains one is a model of the dynamics and one a model of the reward distribution and both of them are modeled as gaussian distributions conditioned on on neural networks so these are neural networks θ being the neural network parameters and this introduces a latent state now into the environment namely the neural network parameters so um what was x before in the inactive inference is now here θ the only thing that the agent cannot directly observe which is what are the optimal parameters that describe what is currently going on in the environment what the agent is currently observing and the second thing we have to do is to we have to make the agent desire a high reward u and we do this here by making the reward part of the observation so now it's observing the the environment state and the reward and then we are setting this u desired distribution in a way that it prefers high rewards namely by setting it to the exponential of a scaling factor of β times the reward at time step t and if we do this we end up the with the following planning objectives so this is the expected free energy now but uh everything puts into its place u and so the planning objective of the agent is now according uh to the expected free energy to maximize this objective here which consists for one of the extrinsic reward which is just the sum of rewards the agent expects to encounter by taking some policy π and then there's also this intrinsic term again which in this case here becomes the mutual formation between the neural network parameters θ and the observation the agent is making so ideally the agent wants to make observations that are both good in terms of reward but also have a lot of information about what the ideal model parameters are going to be uh i just said that exactly and now we can do the use this to forge an algorithm out of that and it it works like this so we take so in every episode we start by resetting the agent to some initial state and then in every step we have to solve this planning objective of selecting a sequence of actions that minimize the objective function i showed here in the previous slide we execute this action obtain a new state and reward we store all of that in the replay buffer and then after the entire episode is done we use this data in the replay buffer to perform the infrared step of of learning our models and adapting them to the data we saw now the challenging part about about this is to perform this optimization here and the first challenge here is already to compute this term even for for any given action so computing the expected reward for for a given action is has been done before in many works and usually here monte carlo methods are used so um just explain estimating this expectation via monte carlo but this intrinsic term which is the mutual information is known to be fairly hard to compute

and so there have been a variety of methods proposed to compute or to approximate mutual information many of them rely on variational inference or amortization but the issue is that we would have to do variation inference over over the model parameter θ which is a fairly high dimensional vector and also we have to do this potentially thousands of times in real time while we are yeah while we're optimizing this function here and so this is simply too expensive to do anything fancy like that so what we are left with is a nested monte carlo approximation and nested because we have an expectation of a curl divergence this means we have an outer expectation and an inner expectation and we we apply monte carlo for both of them the good thing about this is first of all it's very fast and it allows us to represent p of θ by a set of particles because we just we now only need samples so we can also just keep these samples and treat the entire model as an ensemble but the disadvantage is that we require quite a lot of samples of data in order to to get anything done here because um we have this outer and inner loop for every sample that we draw from for this outer estimator here for every example of θ we need to have n_j samples of this inner estimator here of θ again and this means you can quickly calculate if we take five samples on this outer estimator and five on the inner estimator we already end up with 30 samples of θ so we have in like a yeah and a quadratic growth in in samples that we need in order to compute the sum is this this mutual information approximation here and what is also important to note is that each of these samples is a full neural network that needs to be trained and maintained so this is simply going to be very very expensive very quick and so what we propose to do instead is something that is a bit illegal in terms of math or at least it will lose us a lot of formal guarantees and that is to use the same samples on the outer estimator as we used in the inner estimator uh vice versa so to use in the inner estimator the same samples of the outer estimator that means we draw a bunch of samples once and then um we use all of these samples except the one we just now used in the outer estimator to perform this inner estimation and again as i said we lose some formal guarantees because what usually is assumed is that these samples are iid this is not the case anymore right now but we found empirically that this actually improves sample efficiency a lot so i mean here you see a little comparison we did on a discrete like a randomly generated discrete probability distribution where we could compute mutual information uh accurately here you see a plot of the error of this sample reusing estimator the second one compared to the vanilla estimator that doesn't reuse samples and this is over the number of samples so the sample efficiency is actually much higher even though we are losing these formal guarantees here now this means we now know how to approximate or get like at least

some approximation of this of this objective but what is still unclear is how do we actually maximize this objective like what kind of optimization algorithm we are using and the first thing we tried here was to simply use a vanilla cross entropy method this is a fairly common choice in robotics to use this method because it's very robust and doesn't require gradients in anything and it's also considerably fast it works in a way that you initialize some gaussian parameters um so you you always maintain this gaussian distribution of your current action distribution you initialize it to mean zero and one variance then you sample a bunch of action trajectories from this distribution for each of these you evaluate the reward and then you collect the end samples that have the highest reward and for those you fit the parameters of the current gaussian distribution you have over over actions in order to yeah to gradually shift this distribution towards areas of f that have a high reward and if you iterate this for a while you usually end up with a fairly fairly good plan now the issue is that this applying this delivers fairly poor results and the reason for this we quickly found is something that is called detachment or has been called detachment in prior work i should say i don't think this is an established uh term yet um and the problem works a bit like this consider you are in this completely reward free environment right the only thing that there is is intrinsic reward like that leaves you out to explore this and you start in the middle here of this maze and you can either go left or right and in green is everything that is uh intrinsic reward so everything that is not explored has a lot of intrinsic reward left and so your agent decides maybe to go for the left side first it explores it for a while but at some point decides that now the more immediate intrinsic reward is towards the right side right because this part has been explored now so on the right side it's easier to get intrinsic reward so it might switch over to exploring the right side first and remember this is distorted episodic after a couple of steps this agent always gets reset to the middle and has to do this over again and so at some point the writers may be completely explored or whatever and now we are in a bit of a tricky situation because we're again starting in the middle but the entire right set is explored and the left side is so much explored that in the immediate vicinity of this state there is actually no intrinsic reward left and because many planners and many planners that work well on high dimensional problems rely on very local optimization we found that um that these planners are unable to find now a trajectory that leads all the way into this into this zone with high intrinsic reward on the left and the reason simply being that in cross entropy method in order to reach this zone you would have to randomly sample a complete trajectory that goes all the way around this maze and ends up somewhere here in the middle or like at least somewhere close to the border of of

the intrinsic reward here and this is usually not happening the solution for this we found is to realize first of all that the only reason there is no intrinsic robot left in this area here is because we already explored it so at some point in the past we must have taken a trajectory that led us all the way around towards the border here and stopped somewhere here and so what we can do is we can simply remember all the past trajectories we took and use those as an initialization for the cross entropy method to start off the planning from so the way this looks like is that in the end instead of simply returning the plan we store the entire plan together with the current state in a memory buffer and then when we plan for the next time when we start the planning we not only sample from this initial distribution we also sampled from this memory buffer take into account all the previous plans we had and start optimizing those as well and we found that this usually leads to behavior that is very well able to escape this area of low intrinsic reward yeah and that brings me to the experiments we did so this is a task we designed to be specifically hard to explore so the task is as i said before to move this ball into this target zone the tricky thing is that first of all the table is tilted if the robot loses the ball it ends up somewhere here in the bottom and it cannot be recovered anymore so the robot has to wait for the end of the episode to continue exploring and also the reward is completely sparse meaning the reward is zero everywhere except for if the ball is in this target zone and this means that the robot has to explore this entire environment purely based on intrinsic reward and without any extrinsic signal in the beginning until it discovers this reward for the first time so if we apply just like as i told you before this classical technique and reinforcement learning of just applying some random noise we can see that the ball gets dropped immediately every time and even after like 5 000 episodes will soon see we are not able to reach even one third of this table so here on the right is a histogram of positions the ball has visited so far and we are completely unable to leave the start of this table however if we use our method we can see that we actually explore this environment in a very systematic manner and achieve a very good state coverage or at least ball position coverage and our agent is quickly able to find this reward here and find a strategy that consistently pushes it into the right location this is also reflected in a learning curve here's a comparison with a bunch of baseline methods we ran one is pets which is just a model based reinforcement learning algorithm there's also suck and mbpo and neither of these methods manage to to find the reward in the given time uh so we wrapped it up a little bit made it this task a little bit harder so what you see here are um are sorry holes in this uh in this table so now this becomes amazed there's still no uh no extrinsic reward in the lower area and also up until here um so that means that again based purely on

intrinsic report the agent has to learn to maneuver this wall like this is a fairly tricky task right to maneuver this ball around the corners of of this maze into the target location um and although it takes a bit longer i longer to train here obviously because this task is much much harder oh it's also important to know once the ball is inside of this hole is lost and it cannot be recovered although this is much harder and it takes a bit longer you can see that our method is able to to solve this problem and push the ball into the target location at the end so this is a bit of a slower version of of this policy and you can see that it is actually fairly tricky to maneuver around these corners there's a learning curve here unsurprisingly the baselines did not manage to solve this task but i mean this expected since it's harder than the previous one and finally we also evaluate our system or our algorithm on the real system uh so we built this actually in reality and trained it from scratch so there's no transfer going on no pre-training and simulation or anything and you can see here the behavior is similar the agent starts exploring the environment more and more pushing the ball systematically around and uh ultimately um discovering the reward on the top of the table and then finding a consistent strategy of moving it up there's a learning curve for this as well but yeah this brings me to my conclusion so we presented an algorithm based on active inference that is able to solve very challenging sparse reward manipulation tasks and at the core lies this augmented cost function that is derived from the expected free energy and this one like encourages the agent to perform very directed exploration while also maximizing reward we demonstrate that this method works not only in simulation but also on a real system and so to give you a bit of an outlook uh i think what is very important to note is that if we compare to to human haptic exploration we humans rely very heavily on tactile sensing for all of this exploration we have developed a variety of different strategies for for actively perceiving using tactile sensing for example like rubbing an object to figure out the texture or what you can see soon in this video like grabbing it with our full hand in order to to figure out the shape can measure temperature there's all kind of stuff we can perceive actively from objects using tactile sensing tactile sensing is also something that is being actively researched in in the robotics community there have been a variety of new tactile sensors coming out lately one of them is the so-called digit sensor you can see it here and it works in a way that it has a gel that deforms upon contact with something there's a camera in the back and some leds and if there's some contact you can see the deformation of the gel in the camera which is giving you some kind of like local tactile feedback and i think it would be very interesting to see what we can achieve with active inference on using tactile sensors although there is a variety of challenges that need to be solved i mean this is a high

dimensional image the contact dynamics are much more complicated and yeah there's a lot to consider but i think this would be a very very interesting avenue for future research and yeah this also brings me to the end of my talk i want to thank you a lot for your attention i want to thank my collaborators as well boris georgia hani and jan peters if you want you can check out our paper here under this qr code there is also a project page which will soon contain an extended version of this paper that has been accepted to iros but i haven't uploaded it yet so i will do this within the next one or two weeks so if you're interested in more details i recommend you to check this out but again it's not up yet and yeah there's my contact details in case you have any more questions so um if i got that correctly before we do the questions afterwards in the round table is that correct correct thank you very much tim you may exit and re-enter the stream when we go to the round table okay thank you very much and there will be just a few seconds of a break and then we will be going to the next talk which is tim verbellen robots modeling the world from pixels using deep active inference so we'll be right back hello all right here's tim verbellen's talk robots modeling the world from pixels using deep active inference thank you so unfortunately i could not make it live but i regarded this talk for you and i hope you'll find it interesting i'm going to highlight a bit of our research like sketching out the picture what we work on and put it short basically what you want to do is you have don't have robots that can model the world they live in from pixels but basically we can extend it to any sensing modality uh using the factory entrance so why do we work with robots well basically we figured out that if we want to have something intelligent to build something intelligent that is actually doing something relevant in the real world then you need to do action you need to interact with the world you need some kind of embodiment and that's why we work uh with all kinds of robots we have manipulators that can scatter around grouse projects and directed objects we also have navigating robots that can drive around and fly around so you might be wondering yeah why are you looking at both navigation and manipulation and then everything there whereas all over the world there are labs that are dedicated on like manipulation as like a single domain and navigation as another domain and they're like distinct like problems well the thing is that if we take it from another approach if you look at it from an active inference perspective then it's basically the same thing you're doing the same thing and that's why we're trying to combine all kinds of robotic problems from this single scope and see to what extent we can kind of address it and solve them and and to compare where are we compared to state of the art in robotics let's say that's like our overall objective so in this talk i will talk a bit on uh sketch out what is the general approach we're

taking and then some current results in both navigation manipulation and some of the work in progress stuff that we're excited about and working on right so let's start off with what is active uh i guess most of you will know already but just to set the scene and grow for at least next 40 minutes on the same page regarding location and what do we mean with the terminology uh so basically active inference just means to us we have our agent or brain that just built a genetic model of its environment which we call the joint probability over outcome so actions a and then some hidden states z so your agent is distinct from its environment it can interact with the environment by doing actions it gets some observation then i try to model with a hidden state like how is this environment generating me generating my observations and can i model this through these mistakes we use the the single principle of uh optimizing this this this uh this thing by minimizing circle figurity which is like another bounce on surprise or a prediction error and importantly not only you use this to model the system to build a model of the world but you also use this to select the actions that you will hold that will minimize your expected ability in the future and looking at everything from this angle of minimizing your prediction error not only for the pulse but only for the future this basically gives you a very powerful mechanism to start doing robotics just some mods uh equations uh that's that kind of relates to this this reality concept so basically minimizing the energy entails having this kind of complexity minimization so you want to have the simplest model that explains your world where on the other hand you also want to have an accurate description uh of how the world works so you want to predict your outcomes from your states this basically means that you're having all the information that's out there in the world you can actually have this in the state representation but at the same time you want to have your explanation as simple as possible according to fire and this is the list it's basically the variational uh i was here which is like saying i want to be able to reverse the model like if i have some observations i won't be able to infer what is the most likely state that could explain whatever happened to now and then if you look at the future then of course we don't know the outcomes that will come in the future so now it's basically same formulation you can have like the expectation also considers what might happen in the future so what outcomes might i create and then your your expected strategy becomes again like two terms one is the instrumental value it's just says how do i think the outcomes that i witness will actually realize what i expect to be what i want to be like it's like realizing uh preferences and in a reinforced learning context you might see this attack reward symbol but rather having it's coming from the environment it's now more like something that's uh intrinsic that defines what you are as an organism and then is like um gain

like how do i think my beliefs will shift for what i think now that will happen versus what i think my state belief will be if i supposedly i get these outcomes and you're basically searching for the observations that will give you more information about uh what you're doing automatically you get this on the one hand gold direct behavior but on the other hand your your agent is also driven to find out um the development observations find out information in this environment but of course um we have this kind of deep active influence flavor where we basically want to learn this model just by feeding it data basically and we use uh these deep neural nets that you you know uh from from the so-called artificial intelligence nowadays uh like it's basically a functional approximator where you give it some data and then you can you can match any function with them and we use these then to uh approximate the densities that were on the previous slides so how does it go we have our observations or actions we basically feed those to a neural net which then represents this approximate posterior so this outputs like means and standard deviations of gaussian distributions and these are there like your posterior distribution so it looks like a bunch of multivariate reductions and this is then your your state representation like what's happening right now given the observations i had until now then we have a new an additional neural nets that then like predicts the dynamics in the slaving stages like if i do these actions how will my state evolve in the future this now puts again uh parameters of uh gaussian institutions for example and this allows me to plan in this latent state space like what will happen if i do this or that and in order to kind of train them all from that then tries to predict the observations that you expect to see from the state and so the the trading mechanism is then minimizing the energy which entails it here you have your accuracy minimization so you minimize your constriction error but at the same time you minimize complexity with which basically boils down to the l divergence this thing and this thing is like what i expect to happen uh without observations i wanted to have it so as close as possible to what i expect has happened given that i saw the observations and by minimizing the scale divergence you basically of course small to have like very concise um encoding that allows you to imagine what might happen and at the same time emerging information coming from your outcomes but not more than as necessary so you don't want to necessarily model all the details in your in your pixels for example as long as you have enough information to predict uh sufficiently so that's like roughly at the approach and so now we look at some instances uh in in robotics cases and we'll start with the navigation so in this case we have roles like these you know but then with some additional sensors mounted on top of them so we can get get all kind of inputs we just drive them around this map environment where we have rex that

simulate kind of a warehouse setup and then we just train these kind of models that just have to predict for example from camera images what will i see if i do certain actions and then after training such a neural nets then this is the kind of thing that we get so these are basically imagined views from uh from a viewpoint from from inside this lab environment and now you can say okay what do you think will happen if i do certain and this is the kind of thing that that it starts imagining and these are four different samples from this distribution so you'll see at the beginning they're they're all imagine kind of the same thing but as the further you go in the future the more they disperse in their kind of modeling all kind of scenarios what spikes happen in the future and you can also see that even though the model is trained on long sequences it doesn't really capture like long term dependencies or like very accurate uh location information like here if i turn it around i will be facing the wall whereas if i turn it around or just another part of the severe things i'm in the middle of the you know i'm at the edges let's say so it kind of all has this inside the model that these are all kind of scenarios that can happen but because we condition these these these models on the action we can also generate kind of uh action condition samples we can say what will happen so it actually encodes these dynamics inside this model and this is just training from data from just driving around and learning to predict what can happen one of the things that we found out is actually by by training such a model we can then after effects that's that's at runtime compare always your prediction what you think will happen to your posterior belief what's what's actually happened given your observation and then what's the yellow version between the two which is one kind of a notion of of the basis size of the model and if you then put like a new object like you have this stable header there and if the robot drives over it then this kind of dynamic dynamics is it has never seen before which should result in like a huge spike in spike and the robot is like yeah what's going on here and in order to show that this is not just a measuring the difference in pixels we also have this scenario where there are people walking by the robots which also happens when you're recording the data so this is kind of an environment scenario and then actually the robot is like oh yeah this is something that can happen i see some next is nothing to be surprised about so here you can see that it's more than just learning pixel dynamics it actually captures like this is a normal state versus this is a weird kind of dynamics and one of the nice things from from using these learned models is that you can actually put in any sensor so we focus a lot on the pixels because that's also nice to visualize and you can relate to how it looks like but we can also give it like a bigger uh lights you can see the lighter computer sweep and there's some ranging and here we have like a rainbow

which is a bit more expensive basically i mean the range axis like something it just means is something going towards me or is it going further for me and also here we can then have like imaginations of the model like what's going to happen in this state space you already saw the pixels it also relates like okay this is how lighter scans will be this is how radars will be if i if i'm looking at the air i see much more reflections in the radar damage stairs if it looks at the air it has like only a few uh reflections so it actually captures all these dynamics from any of these modalities in this in this latent space so that's one of the strengths of this work but as i told you it it has very a very narrow like temporal depth that can predict for a few seconds maybe one second maybe up to two seconds but then it becomes dispersed and blurry so you cannot really use it for like long-term planning so we figured out yeah what do we need to do in order to use the system but then like do something more uh relevant longer and we figured out that what we need is go a little higher and make like an article model that instead of predicting like the next sensory observation it can like predict uh which kind of state you expect like a bit further in the future and then we found in this kind of model where the blue board is actually what you saw right now but then the red part has cycle where we kind of combine some some ideas from engineering from slab right what is the thing that you expect to be like in minutes or a few seconds there and it's actually like okay i'll do that at a certain location right now and in a few seconds i'll be at a different location and this different location will there be more the kinds of observations and poses that i expect to be there so if we implement such a model it basically boils down to having on the one hand abstraction of the sensory inputs which is what you you saw before and then together some representation of the pose so if you turn around you keep track of it so you know kind of uh where you're heading and you have some sense of fault integration and then these two combined then build kind of a map which is then becomes your your model of the world is then traversing this fabrics and then we get something like this so here we go it's set up uh so here we have the camera now we have a late encryption nation from the sensory inputs here we have the odometry which is very nice you can see it go over the place but then by integrating both like this rough sense of where you are together with uh what you think it looks like and then combining like if this pose and you is pretty much the same as what i visited before then it's probably the same location and you merge the two and so then you see that it actually covers different nails in the um in the lab and it actually makes sense out of it and it and then you can basically use this model also predictive what is this looking like and then you can use this to like plan longer in longer term things so then we go to the more manipulation kind of uh use cases where we have like a robot arm and we didn't

have a camera in the in on the wrist so it's actually getting information what is currently looking at you might think yeah this is like a very different use case but again the way we approach it it's just an agent that can move around and its main objective is to predict what will i move around and then use that to kind of figure out how this first world works but of course now it has much more degrees of freedom but other than that the concept stays the same here we basically have been similar thing where it basically learns other viewpoints from the information government so first we give it back very it's straight looking at the enemy board the table twitter there's nothing and then it's regal's work for all kinds of business like instructions for all kinds of poses in the workspace which just figured out and it was trained on like simple cases with some blocks in front of it in simulation then if you start adding observations to the system you can see how all the different poses are kind of imagining ah there was a yellow cylinder there so in these few points i probably need to imagine a cylinder and this way you can kind of start building the world and reconstructing the world from all kinds of viewpoints um so it's always predicting the future observations from the information it got until then and the model is improving as information but then you can of course also use the model to kind of assess which would be a good viewpoint that i haven't seen before that gives me some new information and then basically what we do is uh we use our this expected energy term to then uh drive the and what we found is that by just doing this mechanism in this case we also give it like a preferred observation like this uh blue cube uh the current observation doesn't looking at that in space i want to found it a bit is actually it first goes all the way up because that it's what it figures out we'll give it more information and then it will start zooming in and scavenge for the preferred observation and then stay there it was a really cool effect to see from just this principle that it works right now it goes up to to have an overview and then once it's uh explored the workspace you can then kind of vary it in normal viewpoint so this is the imagination state of the robot arm wandering around like we had in the navigation scheme but now with the arm action stage so it's again the same principle but we get some similar results then of course we were thinking okay we need to throw a lot of train data like different scenes of different objects added before it could kind of start imagining these kind of scenes but if we look at how we learn how the world works how the world around us is then then bubblers are actually looking at objects and and and manipulating them and looking at a single object at the time and that's how they learn to figure out this is the better way to to learn in scenario how the world is working so that's then what we did instead of like having uh trajectories of random scenes we just made the robot look at particular

objects and then predicts other viewpoints that still from that same object so it's always looking at kind of the same location and predicting how will this thing look like from this side you basically get something where you can imagine for this particular object this is how the dynamics are for this object and then we take a similar model but then we trade it on a different object so now instead of having a huge model that has to learn about any object around we can basically compress this into a very uh well very it's maybe a little brave with a much smaller real net um and then we instantiate a new instance for every new object that we created so we can give it max our a can or a banana and it can start imagining how all these objects look like from other views but then again we can give it like a preferred view and then all of our air which would be a trajectory that brings you to the preferred view and that's then uh that is what you can see here so then the top view is like like the target you just randomly give it a viewpoint and it figures out how to move in this space just by imagining how um what this kind of objects you'd like from from different views then you can also use this to kind of specify objects because if we if we then have a a random object that it either had seen or never saw before it can use all the models and just very like which which model is actually matching my prediction for what i'm seeing right now and this then gives you like an idea like it showed me that object and you can see here that for an initial view it's already pretty good at knowing what it is but in some cases you have like a very ambiguous viewpoint you cannot really distinguish the two objects and then you uh you can just have another view and that will then resolve the ambiguity and that's exactly exactly what you see here so the energy agent will look for the most informative view and then for the node objects it reaches like 100 accuracy and then and for this particular case it's not 100 accurate and why is it because there are objects that might be very related so for example if no it knows it knows about spoons and then you give it a fork and it's like yeah it's kind of looking like a spoon so i'm i'm i'm not sure but and these are the errors it makes but these are like sensitive because actually the dynamics of the objects are very much matching an object i know before and then if you query it like what would be a the next view you would like to see you typically want one with a low expected free energy but if we ask it like what would be a view that you don't want to see then you get this kind of uh uh imagination in the in the the rightmost flow where it's not like foreign or this particular view it's very dark so i cannot really distinguish it from anything and the mystery bubble in bottom right like yeah this is kind of looking like a banana so this is not the viewpoint i'll i'll choose to go next but in practice we saw because these objects are so distinct and a random agent is also good at just integrating information because the chances

are low that you actually end up in this ambiguous viewpoint so in this scenario it's equally well just randomly looking around but the more your uh your objects become ambiguous the more benefit you have from actively sampling uh a particular viewpoint yeah for sure yeah exactly so so immediate like the same thing as as using radar and camera for navigation it's like using textile visual information for for just inferring what this object is like to be uh as a as a robot we haven't done a particular experience yet on on tactile because it's also a bit more difficult to kind of uh yeah just to simulate and and but yeah it's on our minds to do this kind of uh things uh but yeah it's a very interesting route to take a and so finally uh what we can also do then uh is look at a c with different objects and then say okay i want to i'm going to open the gang so i need to have the top view this type of circular gray thing inside the gold image search the only thing it will do is like randomly search around he doesn't see something alive and then go to the target but in our case you know the the system like this thing is probably more related to the can rather than instead of the sugar box so it will draw its attention to the can and then also yeah this is more like a top view and this is more like a side view right so it can imagine like the the thing it should the the movement it should make to go to the top queue so this is exactly then for it directs its attention to the correct object and immediately finds out the movement it should make to kind of get to a similar um so that's kind of the real powerful uh system and you can see this kind of like adding another hierarchical layer on top it's similar to the the navigation where you where you then have like an abstraction of locations now you have like okay i have an abstraction of objects being somewhere spatially arranged then i kind of infer where they are and how it should move in this space in order to operate so then to end something that reflectively and actually get reselectively working on this is basically okay we now have this system where we can build a model where we can relate action to what is happening but one of the difficult thing is still like further the further in time you used to the your your your potential of trajectories explodes it's like yeah you cannot plan using all these these fine grain actions so can we figure out like new sensible actions or skills that we that we can use to explore and so one thing in order to avoid this explosion of options is to amortize a policy to now betray again a function that is then more kind of habit policy that given the state gives you the action but instead of using a reward signal as done in reinforcement learning you basically give it an uh an objective that's more like related to the expected free energy so in in particular here we use what we call latent asian surprise which was like terminally invented to relate more to the oral community but it is just like the information game on your expected states basically and then we found that it

actually found out to play these video games just purely from an intrinsic motivation and also if it compares to other like terms that they add in the url 18s to get this this intrinsic drive we actually found that our method was either on par or slightly better than others and especially in cases where we had some noise in the states so if the observations are kind of noisy or ambiguous the more methods still up then it was it was more easy to outperform the others because they were like okay the noise is like very interesting so we're dropping noise observations although maybe they they don't give you extra information so that was um something interesting before and now we're actually expanding that like instead of having this is more like having an exploration policy but now we're thinking more like okay if we want to plan maybe we just won't have like a few distinct options that make sense to explore like can we can we kind of find some elementary skills that are worthwhile to explore so in this case we have again a robotic r and here we need uh we again explore using this this uh intrinsic drive for information gain but then we give it like 32 potential slots to learn policies and just have to find out which ones are uh good trajectories that in total allow me to explore the state space together and then we ended up with like a manageable set of skills that we can then find for particular tasks uh and that's actually what the some of the things that we're working on right now so with that uh i thought thank you all for listening i hope you enjoyed it should you have any questions you can always reach out via email or via twitter and hope to see you soon the next location thanks all right that was tim verbellen's presentation using deep active inference the next presentation is going to be by ben white artificial empathy active inference and collective intelligence hello everybody my name's ben white i'm a first year phd philosophy student at the university of sussex and i want to start by saying a really big thank you to the organizers for inviting me here today to share this research with you it's a real pleasure to get to do this even if it's just in a pre-recorded format um i want to say right now at the outset that this is very much a collaborative and ongoing project it's work that i've been doing with mark miller and daphne demacus and it's very closely related to the kinds of things that i do here at sussex so i'm interested in looking at the relationship between human human well-being and material environments i'm interested in things like ambient smart technology social media augmented and virtual reality and affective computing which is what i'm going to be talking about today and because this is an active infant symposium i'm going to afford myself the luxury of not going over the basics of the framework and instead i'm going to jump straight in and i'm going to tell a fairly broad stroke story about how we think active inference might be able to shake things up in affective computing so affective computing is a research

program which as many of you will know uh aims to build computing devices uh capable of interacting with human users on an emotional level by uh identifying categorizing and responding appropriately uh to emotions in human users and just to give some examples we have the paper in the top right from the mit computing lab and this project aimed to put affective interfaces into cars because any of us who drive know that there are certain emotional states we can sometimes find ourselves in that probably hinder our decision making progress sorry decision making process and that can have some pretty negative outcomes and so this project was really geared towards increasing road safety by intervening uh in the emotional states of drivers the two devices at the bottom so the screen on camera in the bottom center and the rectangular headed guy who's yellow these are jibo and robot respectively and these are therapeutic interventions that use facial recognition technology emotion recognition technology learn about their users and then make suggestions for certain tasks or games or even clinical interventions that are geared towards supporting the emotional well-being of the user but as you can imagine these are not the dominant deployments of affective computing we mostly find affective computing now in industries like uh recruitment and marketing so for example uh companies like unilever and many many others use emotion recognition devices in their hiring process in order to analyze certain non-verbal responses and facial expressions which they say are indicative of certain desirable or undesirable uh character traits relevant to the job however um you won't be surprised to know that this has come in for fairly heavy criticism so there are certain kinds of worries about this technology um the first worry is that it is simply not functional uh that it's based on bad science and that it doesn't do what it's supposed to do and closely related to that to that is the um various kinds of ethical concerns uh mainly that these systems are in danger of propagating certain kinds of biases and prejudices so lisa feldman barrett for example who is a major leading light in affective neuroscience has come out uh fairly heavy against this kind of technology uh she's labeled it neophrenology and said that it's there's simply no way that it can do what the people who make it say it can do and this is because she says it's based on a very outdated theory of emotion which states that humans have six to eight basic emotions um things like anger fear disgust surprise and so on and that these emotions are expressed through sets of facial expressions which are universal uh across different cultures and different contexts and anybody familiar with lisa feldman barrett's work will know exactly what she thinks about that theory of emotion uh she's not a fan of it at all and the ethical concerns um have been raised by ai ethicists like berber bihari who's argued that basically the fact that these systems are trained on very very large data sets mean

that they are inherently conservative and this is why they propagate certain kinds of biases the two pictures of the hand holding a device you can see on the right hand side in this case it's google's vision cloud in the top picture with the dark-skinned individual that device is identified as a weapon and on the bottom that light-skinned individual it was identified as some kind of electronic device and there are other studies as well that have shown that emotion recognition technology just works completely differently on non-white individuals and so with technology like this making such consequential decisions for people's lives it's really really important that we start to get this right and we don't have these kinds of outcomes that we have in the current program okay so we need the strongest most up-to-date theoretical underpinnings that we can and we think the place to start with that by the place to start in updating the science is to recognize that these devices are very problematically disembodied they're very superficial and they are inactive in the sense that they don't perform any actions and this of course is a million miles away from the way human beings interact socially and emotionally and we've known this for a very long time so we've known for example this example from luis pizza's work that emotions are not simply um partitioned off from the way human beings think or act actually cognition and affect are very very closely intertwined okay there's no discrete separate brain areas that only do emotion and only do cognition furthermore research paradigms in cognitive science so embodied cognition for example tells us that actually we need to go even further than that and recognize that embodied action is an integral constituent part of how we think and feel and then the other ease that make up for e cognitive science so in activism activism extended cognition and embedded cognition have come together with niche construction theory in really interesting ways to tell us that actually we also need to consider how elements of our external environments can scaffold the way that we think and feel this is where active inference comes in because we think that if we're serious about designing these kinds of devices taking into account these most up-to-date theoretical developments then active inference basically brings all of that in with it so active inference is a theory that really elegantly kind of intertwines action perception thinking um cognition and affect together under the unified imperative of minimizing an agent's surprise so that's the first thing that it gives us off the bat it gives us this unified computational unified conceptual framework that can be shared by researchers in different fields but mainly as the name suggests active inference really puts action front and center it's not some afterthought or just some kind of bolt-on gimmick and i think it's worth reflecting for a second just on how central actions are to the way that we interact socially and emotionally okay so human beings are very

far from being a passive uh classification device we're constantly sampling the world and probing the world in order to get more information so that we can update our models of the world so imagine the following scenario which happens to me fairly often uh imagine you're on public transport and somebody's giving you a weird look or maybe they're scowling at you in some way we tend to not sit there and just look at their facial expression to try and work out what's going on we have other avenues um open to us so we might look behind us to see if they're actually looking at someone else we will probably look at the broader scene for some context to see if there's something going on that can tell us more about that scowl or depending on our mood we might even scour back at them or flash them a smile and see how they respond to that and i think another scenario that really brings this intuition out very strongly is to think about the peculiar kind of tension that comes in a job interview so i think that tension is the result of a confluence of two things firstly very high uncertainty uncertainty that's really important to us and also the fact that our usual embodied epistemic resources have been straightjacketed by social convention because it's the case that in a job interview even though we want to know a lot about what the other people are thinking and feeling social convention dictates that we can't ask them we can't prod them and we can't really sample the scene in ways that are going to give us more information we're kind of stuck to the chair we just have to wait and see and that's an unusual situation to being because so much of uh human social and emotional interaction relies on active states to use active inference terms so speaking listening prodding smiling scowling raising an eyebrow all of these things are embodied that we take to learn more about social setting and i think the thing to emphasize is how important context is as well so the ability to kind of actively survey a scene to drink in context but also the way we learn about the relevance of that context is um something that's built up and scaffolded through action and uh different kind of patterns of practice so if you think about the scala on the bus again there's a high degree of uncertainty around that facial expression but if you imagine that same facial expression transported into the onto the face of somebody on the other side of a boxing ring all of a sudden the uncertainty around that facial expression is minimized because the context of boxing ring tells you everything you need to know about why that person is scowling at you and this emphasis on context is something that's badly missing um in current iterations and the active inference community is already producing really cool work premised on these kinds of insights so there's this paper thinking through other minds by samuel vasiere and colleagues and this paper highlights just how it is that we come to understand our socio-cultural niches through precisely this kind of active

social foraging so it's really emphasizing the importance of context and the fact that we come to learn about context through action and there's an example um of where we see a gap between artificial systems and humans when dealing with context is how artificial systems and humans compare when performing selective attention to in regards to some tasks so it's a really pervasive problem in artificial systems that they don't tend to look at the same places that human beings do when human beings are surveying a scene for some kind of task relevant information so the question is how do we get artificial systems to drink in context in the same way that humans do and then how will that improve the performance of emotion recognition devices so selective attention is all about filling in epistemic gaps it's about filling in gaps in your knowledge with information from a scene that may or may not be task relevant one of the reasons humans are so good at this is because obviously we have this huge knowledge base of uh what different contexts mean we live in the world we've always inhabited socio-cultural niches and so we have a lot of experience but as i said before it's important to emphasize that the way that we learn about context is about sampling different contexts it's about the fact that we have our entire lives been actors in the world not passive observers and i think this work by Moser and colleagues that you can see on the slide it's really interesting because it shows that active inference is capable of modeling selective attention in ways that give us much more human-like results so they used certain kinds of internal precision dynamics and they demonstrated that these precision dynamics can map accurately covertly um task relevant and task irrelevant features of a scene and then update precision estimates in in relation to that information which then drives overt actions which then serve to update the system's model so it's this very close relationship between covert attention and overt tension which i think is really interesting on this account and i think active inference is a really powerful framework for recognizing addressing context generally and for the importance of action in learning about context so the models that i just outlined they provide the tools for this very elegant top-down first principles approach to selective attention which is based on these internal precision weighting dynamics but also on uh embodied action perception cycles and of course one nice side of side effect is this side effect of this is that systems based on this would be able to autonomously select the data from a scene which is going to give them the most epistemic payback and this means that they can do away with the very very large data sets and long training times which ai ethicists have said are probably the root cause of a lot of the ethical concerns that i talked about earlier and so from a practical standpoint this means that we need to think about building affective computing devices which are not merely inner lumps of

plastic we need to start thinking about approximating something much more like a fully embodied agent one consequence that's really fascinating about the kind of active social learning that scaffolding scaffolded through other minds um that i was talking about earlier with the thinking through other minds paper is that active inference agents can come to enjoy a degree of synchrony between their internal states this paper by carl fristen and chris fifth a duet for one it utilized simulations of songbirds to show that quote generalized synchrony is an emergent property of coupling active inference systems that are attempting to predict one another so in rough terms what they demonstrate is that according to active inference meaningful communication between two agents requires that they are sufficiently able to model one another in a kind of infinite regress so what it is is me modeling you modeling me modeling you and that by doing this by making and testing these kinds of predictions we essentially ultimately converge on model model synchronization and this is a core part of the paper the original paper by daphne de mekas that she did with um fristen and parr which has already suggested that if we take active inference as a starting point for building affective computing devices then what we have is the prospect of an artificial system which can potentially sync internal states with the user and that's obviously going to hold an awful lot of promise for um certain applications of effective computing and i would say to anybody interested in the kinds of things i'm talking about now to go and start with this paper by daphne because i think it's a really interesting and wonderful starting point one thing that we want to say is that for this sort of deep affective synchrony between artificial devices and users it means that the artifact itself will need to have some kind of interoceptive signals some kind of internal affective dynamics of its own and it needs to be able to act in ways that expresses those signals so so far i've been talking about acting in ways to express those signals and acting in ways to sample the but i want to say something now about the prospect of active inference devices which actually have their own internal affective dynamics because i think that the active inference framework has already shown the potential to provide this what i'm talking about here is some fairly recent developments in the framework called aerodynamics um and using aerodynamics we can start to understand how embodied affective states are an intrinsic part of the motivational drive for curiosity and epistemic foraging so one of the really elegant famous strengths of active inference is that it has the power to dissolve this opposition between explore and exploit and while there are there have been numerous strategies for building the motivation to explore into artificial systems active inference has the potential to put embodied activity and emotion right at the center of solving that problem and so this is

obviously going to be relevant if we want to build emotional recognition devices that are intrinsically motivated to probe the internal states of their users so i think it's worth taking a second to refresh how active inference accounts for emotion and effect the first attempts to understand um interoception and active inference were they beared a lot of resemblance to the way that we were thinking about perception under active inference um so it was about um predicting um signals hidden states in the world except that the signals that the brain was trying to predict were internal signals they were coming from inside their own body um so gastrointestinal respiratory circulatory signals and feelings like hunger thirst temperature pain these were seen as top-down predictions about the hidden causes that underlay those physiological changes but it was the case that researchers that were working in affective neuroscience so people like lisa feldman barrett uh neil seth and mika allen they were quick to add that these interoceptive predictions probably held a kind of special prioritized place in terms of the overall system because they were likely to ground other predictions predictions about the external world in terms of what really matters which fundamentally is maintaining the homeostatic states of one's own body but more recently than this affective states have been hypothesized to fill another role within the active inference framework which is essentially says that felt bodily states things like mood and other affective states valence bodily states they reflect a kind of second order information within the dynamics of the active inference system and that information is essentially tracking the rate at which surprise is being minimized relative to the expectations of the system so according to aerodynamics affective states are essentially uh just the subjective level feedback about how the system is doing at minimizing surprise at keeping itself within expected bounds relative to the expectations that we had going into that scenario the second order information doesn't just it's not superficial in the sense that it just reflects that kind of information but it actually plays an intrinsic role in modulating the internal precision dynamics over action policies so from a phenomenological perspective i think this makes really intuitive sense so when we're doing better than expected at a certain task we tend to gain confidence we might take more risks and when we enter a certain scenario or task with a particular action policy which doesn't work out the way that we expected it to we would be very quick to switch things up and try something else and in this sense aerodynamics can be said to keep agents flexibly attuned to the opportunities for success within their environment as they learn and develop new skills and abilities and the thing to notice is that agents that are outfitted with a sensitivity to aerodynamics and naturally curious because finding new um uh new surprise in the environment which can be successfully

minimized it literally feels good to us um and the places where um we find surprise that we can minimize um in the greatest amount is at the edge of our skills and abilities and this is why we like to find scenarios that are kind of maximally challenging without being frustratingly challenging without being too hard so we like to occupy areas that are neither too well known nor too complex and aerodynamics also plays a role in helping us to direct and enhance learning so um surprise and its reduction rates signal the expectations about the learnability of particular situations so that helps to guide our attention and prioritize certain areas or certain tasks where we know we can find the most success and we've already seen this kind of optimal surprise minimization show up in robots in terms of curiosity so there's this work here by odoya and smith where their robots were trained to seek out optimal levels of complexity where the most learning can take place and specifically now there have been active inference approaches that have begun to use aerodynamics in real world robotics so this is really exciting and this is kind of real world proof of concept in the work of skelassi lara and syria and these researchers have actually built robotic systems that make use of this internal aerodynamics machinery so their work has shown that robots that are equipped with aerodynamics are actually better able to manage uncertainty um by fluidly selecting adaptive actions in an environment compared to more traditional and traditional approaches so artificial agents equipped with internal aerodynamics are better able to learn um and then autonomously select the proper surprise minimization strategies in any given situation and they do this by allowing their valence states that second order information about performance relative to expectation to weigh the selection of the most suitable behavior so in other words by allowing that second order information to have a direct impact on the internal precision dynamics over action and this work showed that this kind of internal aerodynamics also provided a way for artificial agents to navigate the temporal aspects of goal selection so basically what that means is these agents were very knowledgeable about how long they should persevere with a certain task and when they should give up which is obviously something that even human being human beings struggle with a lot of the time thinking in terms of aerodynamics it shows us that affect is intrinsically linked to goal selection and we want to suggest that by introducing these aerodynamics into affective computing devices we would start to see devices that are not only motivated to kind of exhibit a kind of curiosity in implementing new policies for action but we'd actually start to see a real paradigm shift in the affective computing program to a much more biomimetic approach so instead of just having classification devices in lumps of plastic or in smartphones we'd start to see embodied devices that can actively engage with the

world and that have their own internal affective dynamics based on what we think is going on in living systems so this is really exciting and this new wave of affective computing devices would not only um be able to perform much better but we think it might go some way to um addressing some of the ethical concerns that i was talking about earlier but the thing to be really clear with we don't want to a kind of disclaimer at this point is we are certainly not saying that an affective sorry an active inference approach to is a replacement for thinking about all of the kinds of social justice issues that come with the implementation of this technology we are just kind of speculatively saying that it's on first glance it certainly appears like some of the ethical concerns might be addressed by this new approach um but we think there are going to be a lot of benefits of this new approach so first if you think about the kind of model synchrony um that i was talking about earlier and think about that within the context of therapeutic intervention we think that when we get this degree of model synchrony um any any dysfunction in the user's internal dynamics is going to be mirrored in the internal dynamics of the artifact and so this is going to be make the device very well placed to make suggestions about potential interventions and this is essentially what cbt already attempts to do so this will be building on approaches that have already been proven to be effective and the next thing is to think about the fact that so far when we've been talking about action i've been talking essentially about epistemic foraging um but once you have the possibility of humans and artifacts establishing this kind of model synchrony and you have these artifacts that are properly embodied and able to act in the world it might be possible for the artifact to begin to install prior preferences uh about what states in the agent are actually preferable such that the artifacts may actually be able to steer with a degree of autonomy the emotional synchrony between it and its user to specific ends so when we have this uh we have these active inference theories um beginning to emerge of depression and anxiety and other kinds of disorders with that full understanding coupled with the kind of model synchrony i've been talking about we start to open up avenues for the device itself steering the user away from these kinds of dysfunctions now it might be the case that these active inference devices come with their own set of worries and ethical concerns i think it's very plausible that they do and it's something that we're going to be thinking about as we go forward in this research but i don't have time to explore it here but i think the last thing to say is most speculatively is that this active inference approach also sets the stage for beginning to address the well-known value alignment problem between humans and ai devices basically what we have here to our mind a initial and very speculative building blocks of building artificial systems that have a degree of genuine

empathy with their users so these devices would not merely be simulating empathy or passively categorizing human emotion they would generally have genuinely have their own internal dynamics that would synchronize and match with the internal dynamics of a user and this is going to be a bedrock for much more interesting and much more rewarding human ai collaboration into the future and that's the end uh i wish i could be there to answer questions um unfortunately i'm not uh i'm pretty sure mark miller's going to be there with you so maybe he'd be happy to answer some questions but i would also encourage you to get in contact with myself i'd be really happy to hear from anybody that's interested in this kind of stuff um you can see my email there be white at sussex.ac.uk and you can get in touch with me on twitter as well at midnight biscuit thank you again for listening it's been a real pleasure to get to present this thank you very much um should i start we're back and yes please this is the presentation of north said learning agent preferences thanks north for joining and take it away brilliant thank you um daniel um so just before we get started i just wanted to thank daniel and the active inference um conference i guess um for inviting me to give a talk on this project um i'm really excited about that um so just to introduce myself i'm noor i'm a current phd student at the welcome center for human neuroimaging with carl fristen um this is some work that we've been uh thinking about over the last uh year or so feel free to i don't know how well so do we have questions that um sorry do we have questions throughout the presentation or are they at the end um but if there are any questions we'll be uh taking questions in the live chat and then the round table will be bringing them directly during your presentation okay um yeah no worries okay so the project is focused on learning agent preferences and it's from my perspective is super interesting because that sort of changes the dynamics of how you consider the problem setting and that's what i'm going to start off with but before i do that um i just wanted to highlight my wonderful co-authors that i'll be uh sort of presenting the work on behalf of uh so we've got panus alexey zaff lance and carl um and the project so the work i'm presenting is based on two different projects and i'll highlight the the different work as we go through um so my print okay so i'm trying to flip okay let's yeah so the the way the presentation's going to be structured is just i'm going to briefly motivate the the problem setting um and then describe the problem setting in a bit more technical details and then really drill down into exactly how we can learn these preferences that we can equip the agent with and then some experiments and remarks so what i wanted to highlight is that when we learn agent preferences there's usually a bi-directional association between the the agent and the and what i mean by that um is something that you can see in this graphic really clearly so you've

got uh the main part that the agent would have taken um as it was um walking down this particular route uh when hiking but as perhaps many other people are joining along the agent ends up um walking along the shorter or maybe the more smaller part um so what this highlights is that agent preferences are essentially dictated by the environment that it's surrounded by right so depending on the constraints so for example other agents maybe an animal or something else happening on the road would mean that the agent um ends up taking the second part instead of the first one and as a consequence of that it changes the environment so as more and more agents do the same thing um the the shape or the construct of that part becomes more prominent and it becomes part of that environment and what i'm really interested in as part of my work is this sort of bi-directional association between how the agent changes the world and the world changes the agent's preferences because it constrains the the actual state space that it exists in um but as part of this project and the work i'll be presenting we're purely focusing on how the agent's preferences or the how the agents uh i guess objectives are changed as a consequence of the environment constraints um but before i do that i really um i wanted to highlight what preferences actually are because a lot of the time um we're not really aligned on what that means um so i'll just briefly uh describe how we are considering preferences so preferences here are usually a subjective assessment of what agents would like to experience and this can be continuously learned or modified even in the absence of external feedback so there might be some internal motivations or some um objectives that the agent is learning internally that shape exactly how the agent wants to behave in the world so in the previous example when we were looking at the case of going between the two parts depending on the constraint of the environment it's essentially the agent's subjective preference because it could have taken another route which uh so for example maybe here um and that could have also shaped the environment as well um okay um so let's motivate this problem setting slightly more formally um so we're going to be working um as part of um and the idea that these agents that we're interested in have an internal model and this internal model is um composed um really briefly of uh three important components um so three important random variables and then one deterministic variable so let me just qualify that a little bit more so we've got our outcomes so this is something that the agent is exposed to in the environment for example um the the actual i guess um constraint in the instance of the the hiker being having to choose between the two parts for example um it might see a hindrance or it might um like have some grass or um some some other agents that are exposed to and it needs to that's the data coming through and it needs to then identify whether it wants to choose

uh one part or the other part so that would be based on the its own inferences about what that outcome actually means and so that's denoted by s here and then based on that it needs to then decide what action to take u_t and that's denoted by the a here and at the same time the agent is keeping track in this particular formulation of the the agent's model or the general model that we're interested in which is denoted by this deterministic variable h_2 or h_3 depending on which time point we're interested in that is essentially this deterministic recurrent model that we have u_t that's encoding all the prior history u_t the the actual states uh sorry the actions and the states that the agent has been exposed to sorry in the past and that encodes what is the the updates that are then used to select the the posterior estimates uh for the the next time point u_t then you've got your in this particular model you've got your latent state and prior and and we're calculating them in really a specific way so your prior is defined as categorical distribution and this becomes really important for us because this allows us to use some conjugacy rules to update the way the agent's preferences are land and i'll come back to what i mean by preferences in this technical setting a little bit more on the next slide and the state posterior here is again a categorical distribution that the agent is estimating based on the history and the current observation it's been exposed to then we've got the standard formulation if you're working within a mdp formulation where we've got a transition function so this is denoted again as a categorical distribution conditioned on the history of the agent that's what the agent is encoding and then we've got an image predictor u_t that determines exactly what would be the next u_t time point oh sorry what would be the next observation given the history and the u_t so the idea with this gender model is that you have an encoding of how the agent is representing the world it tells you exactly how this outcomes are then u_t inferred as particular states that then allows the agent to evaluate particular actions but i haven't really u_t defined how the actions are selected u_t so we work in a standard uh active inference setting where the are defined as u_t being sampled from some probability distribution a so the actions we saw before which is calculated as the $\arg \max$ of minus g over a so what is that exactly u_t so for our work u_t we were interested in essentially extending the minus the expected free energy u_t with a conjugate prior and so the expected free energy in standard terms would be something where you have an extrinsic imperative you have a salience formulation and you have novelties extrinsic imperative is something that's contained by the environment saliency is when you want to have accurate belief updates and novelty is when you want to be able to estimate your world appropriately given the the parameters of the model that you've been able to learn u_t when we extend it for the preference learning setting

um and this is something other folks have done um maybe introducing it as part of uh prior over the outcome space um so we're looking at as part of a prior over the state space so sorry um let's see if i can get the color to change okay so this is the the prior and we're conditioning it on a categorical distribution d which i highlighted before and that allows us to take him to um i guess i'm sorry use uh some of the conjugation rules of interest so now i'm gonna um go to the next bit so um sorry just going back here so um what do we have so far um before we move on to how do we actually land this um d that we were interested in the previous slide so we have an agent who's equipped to the model the agent is interacting with the world and based on this interaction the way the agent is learning its preferences can shift and the way is shifted is a consequence of the type of actions it's making that we had using the expected free energy so how do we learn um preferences um so in the literature there's been um well at least in the psychology literature and some of the the spiking neural networks and other formulations um there's been a few different ways it's been proposed how agents are learning preferences um so one of them is mere exposure effects so the idea that when you're seeing something quite frequently that that pairing is more absurd sorry more preferred than if you want um seeing that and this can be uh categorized as a heavy plasticity learning rule then you've got attention mechanisms where when you're attending to an option um it becomes more preferable so maybe you're selectively looking at x or y and based on that you've filtered out all the other data that you've been exposed to um and that's what becomes more uh something that you as an agent would prefer and if you were to think about a more i guess um biological construct of that that might be a consequence of some synaptic gating that can encourage the enhancement or some sort of suppression of the data or the noise that you are another formulation could be the contextual effects where an option is only preferred when it's compared to some other options so there's this relative comparison happening and based on um that in particular in certain settings you would want to do x instead of y and here x could be taking the path uh the longer route sorry the wider route route in comparison to the maybe the the tinier route that we saw in the hiker picture and that could be on you you only prefer the wider route when you don't see an animal some blockage there and this encodes some behavior relevant signal selection for the agent um so now what i'm going to do is i'm going to go through some rules um and some formulations that we've been thinking about in terms of encoding preferences and they are um the way we've um sort of privatized the the learning of those preferences it's it is aligned with some of the um i guess things people in psychology have also been thinking about so the first one is learning preferences from your exposure so

we start off with extending the agent's gender model with a conjugate prior over the prior beliefs and what that simply means is that we take the categorical distribution that we've conditioned our prior state on and then we introduce a prior over that um and that allows us to take into account some of the the conjugacy up to update rules that we're interested in um because our categorical distribution is oh sorry our prior distribution over the state space is d which is a categorical distribution we can now define a directional distribution as a conjugate prior which is denoted here and we've just got two different formulations here of how you can update so essentially as you're exposed to more data you are introducing um you're counting your suit accounts over the summation of all the suit accounts in that particular um factor that you you're interested in taking this into account we can do have been learning um rule which we do using online interactions through preference preferences so the way it works is that given your hyper prize in the time point before you can update those based on um some alpha sorry some learning rule alpha and also the belief um updates you've had in the past so this is denoted by the s for that particular pseudo counts or the territory um primatizations uh for this current time point and you add all of this together and that gives you the updated um so the way this particular formulation works is that the more you see something the more you're going to prefer it because there's a very simple learning happening here and in the simulations and the way we formulated at the moment we've got alpha set as a static parameter equal to one so you can manipulate that and the way you uh you can manipulate it where if you have it alpha going to uh greater than one um you will then wait the new data coming in a lot more in comparison to if you had it out for less than one um then you're not taking into account the the weight of the new data coming in as much and what this allows us to have is accumulation of particular contingencies or the way we are privatizing our prior and these dictate the land preferences the next formulation of learning preferences i wanted to speak about was how we can learn by attending to preferred options so here we're going to slightly change the model um by introducing us an additional preference learning component so we again are extending the agency enter model with conjugate price over the prior release so the hyper price so this is exactly the same as before but the the different thing that we're adding is the synaptic gate scene code preferences and these just are computational homologs that we've introduced um in terms of the actual uh biology of like the the formulation i think that's open to interpretation but these allow us to have an attentive mechanism which is what we're interested and we do this to a two-step procedure so first we encode memories and how that how that works is we take a sample of all the data that the interested in um sorry the the agent has actually um okay

sorry um so the way the way the encoding of preferences is working is we have two components um which we're then um combining together the first component is the the online exchange that the agent has had so by online exchange i mean um the agent is selecting its um actions based on the expected free energy and that's allowing you to essentially gather data about all the different trajectories it follows depending on what action is taken and it's got the data and its posterior estimates given that the next um so this would be the the on policy the second bit that we have is the imagined interactions so this is when the agent is offline in the sense that it's no longer being exposed or given outcome with outcomes uh based on its imagined or the way it's interacting with the world and the only thing it's getting is the the updates in the states uh the latent states depending on what actions it's selecting and this is the imagination part so what we're doing is we're combining this together and we take 30 of this and all of this um so these are 10 steps into the future um and then we're essentially interleaving them together and that gives us an encoding of the memory um the reason for only using 30 is to allow for um the imagined interaction of the new percep things that the agent is considering to be taken into account and the the idea with the interleaving is that we're allowing for but the the real experience and the imagine experience so the the reel is here the on policy experience to be um used to shape the way the agent is um encoding its perception of what has happened in the past um and then using the encoded memories we then encode the preferences using um a selective attention process so this memory buffer that we have here is what i showed in the previous slide so this is the memory buffer um and then using this we essentially um encode the preferences using two gating mechanisms so the first one is an attention block and the second one is a gating block the attention block um essentially weights some part of our distribution slightly higher and then the gating block constrains or restricts that data out by filtering it and we are optimizing these two blocks using maxim entropy and the idea optimizing through maximum entropy is to some shifts in what's happening and by shifts i mean to have a more flexible representation because we're trying to maximize the entropy of the distribution okay and this formulation allows us to encode filter contingencies that can dictate land preferences this is a slightly different formulation built on the same bayesian updates but we're introducing this uh selected memory component um okay so i'm going to quickly go through the experiments um so we evaluated this um the two algorithms in a 16 by 16 by 10 grid world um so an example grid is here so the agent is presented with this image including its own location at each time step and we've got four distinct states so we've got red we've got blue we've got the light green and the dark green as well and in this

particular formulation we have no reward or score outcome modality so the agent is learning purely on its motivation to understand the world and if there are questions about that we can talk about it and the grid is changed every k steps and k determines how volatile the environment is and at each episode the agent is initialized in some random application so maybe here or there and that constrains how it interacts so for sake of time I might skip this bit but essentially the tables are highlighting exactly what the training parameters were and they were fairly consistent actually exactly the same between the two algorithms so the pepper formulation where we have had been plasticity and the north formulation we're doing non-reinforced updates using selective attention and then the preference learning parameters that like how long the planning horizon was so it was 15 and we have an episode length of 100 and we do this for 50 episodes and we reset so k is reset every 1 25 50 75 and 100 steps and this gives us a nice way of evaluating what's happening so then the first thing we were interested in was evaluating how the preferences are shifting in a static environment so this is one where we have Hebbian so this is learning and this is where we have the attentive gating measure here and with the heavy and we can see the environment so this is the on the x-axis we have the states dimensions and the y-axis we have the epochs or the time consideration and this is the same for both the figures that we've seen at the moment and we can see with the Hebbian one that as we go further down it becomes more concentrated and there's no really shift here whereas for this for the attentive preference planning measure we've got these random blocks that appear that once they're before but they also disappear so it's quite interesting that over time the preferences are shifting as a different measure compared if you're doing a qualitative assessment of the comparison between the two and then the content and our post talk analysis of trying to actually understand what is happening quantitatively we compared the heavy learning formulation with the attentive selective attention preference formulation and then we compared it with the baseline which is the expected free energy of g and on the y-axis we have the environment volatility again this is the k percent k but just denoted as percent and then we on the y-axis we have preference satisfaction and exploration trade-off that we're using as orthodox distance and this is just evaluating how far particular trajectories have shifted so what we're comparing is whether there's an increase exploration or not depending on which preference metric we're using because based on our qualitative assessment that we saw before there is this shift between the different encoding of preferences given the preference measurement sorry formulation that you're using so we can see that when we get to

50 volatility there is this shift from exploitation to exploration um for the pepper algorithm and so we see that based on this nice mode of the distribution that we're seeing here whereas for the the new uh formulation is slightly um expanded out but it's not as exploitative as the pepper formulation and then when we are looking at extreme ends we can also see that there is this shift from exploration exploration to exploitation um for the pepper formulation but we don't necessarily see it for the more formulation so at the moment there is some quantitative differences in the way the two agents are evaluating encoding preferences and how that shapes the behavior and this is just an example of the north formulation in terms of how much it's exploring the path and the the way it's interacted with the so these are the grid worlds and this is just a heat map of that exploration trajectory um then i'm just going to quickly do some takeaways um so both of the formulations that we have even though quantitatively and qualitatively they are providing different ways of encoding preferences they do have a tendency to influence the agent behavior that's different to the baseline or the expected free energy where you didn't allow for this change uh preference assessment and um if we are to compare with the standard reinforcement learning setting we are here casting what is preferred to an agent instead of the environment or the designer and it's particularly important in a robotic setting where you want to be able to go back and allow the agent or a robot to be able to shape some of its own goals and objectives purely if it's working in a more creative setting maybe in one in settings where it has to do a really specific task this type of formulation might not be the best um but this does provide a flexible formulation um where we're only modifying the the preference learning component and using the same gender model so the the initial general model that i defined for the agent is kept consistent for both the pepper and the normal formulation but the behavior that we're getting is through this additional component that we add but the the key thing to note is that these preferences are a consequence of learning suitable um learning a suitable gender model so if your gender model isn't that good then the way you would learn the preferences themselves might not be the best so there's a slight trade-off um because whoever's designing this formulation or working with this foundation needs to take into account how well the the gender model has been encoded or launched by the gen agent um and on that particular note i'm just going to end the presentation um so thank you so much everyone for listening and i want to thank my quarters uh for this great work and my funders and i've got the the qr codes for both the the papers if you're interested um and that's it thank you awesome talk okay i'll just stop sharing and you can depart the room and you can rejoin for the round table if you'd like otherwise thanks again for joining thank you so much

have a good day bye bye see you bye awesome all right the next talk is going to be by wen hua chen do a control for exploitation and exploration and its applications in robotic autonomous search just one second hi there yo hi one all right we're back we're back with with wen huachen thanks again for joining and please take it away okay thanks for uh you want me to this particular meeting uh it's quite interesting for me and as we are not i'm not always operating this particular community so thanks for providing me this uh chance to share my working experience with you so my talk is about how to uh develop a autonomous search strategy for uh robotics and investing at a context about a chemical biological and other interesting dispersion so the approach i'm there is called due control for exploration exploitation and as you will see is actually a lot of similarity with the active influence theory so i came from uh loughborough university i'm working in the airspace and automotive and union department so i have quite a strong engineering and background but not on the so this is the best outline of my talk and you can i can show you some background about the application and also discuss design method and then talk about the simulation and experiment results and also particularly i'm interested to share with you about my thinking about what are the relationship between the the approach we discussed here and also the active influence reinforcement learning and other similar kind of area and then talk about what is the way move forward so and let's talk about the autonomous search as the case study so this is a so basically they can why they found it in the natural environment and for the the polar bear if they try to find the prey or the food and they need to use a small audience and also similarly you can find insects if they are such mating and other food they can use similar kind of strategy the idea here is and by making use of the sensors they are near to reasoning about where the food or the soil might be then think about it what is the best strategy in order to find it and particularly interesting for us is how to convert this um in kind of intelligence from natural world into the engineering area so we are able to teach robots or uavs try to search and be chemical marriage and resources and also and in the future you can be using for environmental uh force uh enforcement for example and try to fund the protein aware resources and many many other applications so basically you can think about it the idea here is somehow like you try to develop a small dog which can slip around them try to find the drugs and other endangered materials this is not a new area there are lots of research in the area and particularly bio inspired a lot of research people can be using and the campbell texas or and the other reactive strategy like you fly down the wing if you found something you try to follow the trace in order to search the resources another mainstream uh working this area is

based on we call the information theory approach which is uh you treat this kind of process as a information game process somehow at the beginning you don't know where are the sources and you have high level of anxiety then during your search you drive the level also in the lower and lower so this is actually is that you can think about it this is a information game and then you can using the reward function like entropy like the uh k or divergence of many others to measure the success of your search and then based on that you can derive the strategy to driving to do this but now we're not we're looking to another angle and we treated it as a control problem and then think about how to link this with a active interference the same kind of work so when we try to search the uh a source and you don't need any information so basically on the robot you have a chemical or biology agent gas sensors and then based on that every time you need to reason where you need to move your robot in order to have best chance to find the source so there is a strong interaction between the robot and the environment and the environment rover and also they have a strong impact coupling between the the perception and uh and any decision making so basically if you decide to go to different location and then you will take a different kind of chemical sensor maybe this will affect your belief about where the source might be and this will also change your cause of action because based on that you need to decide where you want to move in the next step so there is a strong covering between the perception the planning or action so and this is also a typical example of cycle that you need to trade off between the exploitation exploration so you need to maximize your chance so a lot of people in this field are much familiar with this kind of trailer i will not go into the detail now i will try to explain to you about the strategy uh whether we have development so and now we try to formulate the problem and uh and there are some methods here and you can ignore that but they try to understand the high level understanding and what the the math here is each time we have a x which is a state of the agent of the robot and also you have a variable set of action you want to take and which is basically try to make the robot move forward backwards or left and right to move to different direction with different step size and also you have the moments the measurements in this case is you are first is your gas magnet on the on the sensor you can have chemical sensors on the on the robot and also you have send the moment about your position or your location with respect to the and the other things that we have is the unknown information about the source and also about the environment and which will include the location of their source and also include that constant environment like the wind direction and the speed which will significantly affect the dispersion of the chemicals or any uh audio so and the idea here is you take the connect all the

data during the process and which is included anytime the action you take which is you and also the sensory sensor measurement but which is that here so you add all together give you something we call the information state which is a connection about all the data you have so far and then we decide what is a cost function or reward function it should be and in the simplest way what do you think about okay and my aim is to try to move my robot position close to the source location as close as possible which is quite a sensible uh the mayor but the problem now is i don't know where the source is location resolves it so what is this is why the condition on all the data is connected so far so it conditions this one trying to minimize this cost function which is a typical way and in our control community and there is a particular name called the stochastic mtc model for the control and but what we want to do is move away from this just doing this we want to uh to further and because this will move further this is actually beautiful link with the active inference so what we do is now okay we're not just the conditional on the other data we connect so far but also we connected on some virtual data so what is the basically is we also added the future all the the actions and what are the future outcomes so somewhat because you have a model he said oh if i do this what is going to happen what kind of moment i'm going to get if i do another thing what can moment i'm going to get and how this man will affect my belief of the world of the environment so that it means i now condition on not only on the ik but also condition on ik plus one and here and we need to have taking into account contraction on our future measurements and how future amendments will affect our belief okay but you know before i introduce the uh the detailed mathematics a little more method i try to give you uh some definition of mutation so suppose any time you have the probability density function of any unknown parameter θ you want estimate this is a pdf for θ density function of that and if we take the the mean the expectation of that this will give you something we call the nominal estimation okay use the mean of that as your nominal estimation and you try driving europe maybe to the nominal estimation location as a matter okay but also we have to quantify how the uncertain level of this so how reliable this your estimation about the environment so basically we define the error between them and then the nominal one and from any measurements then this will actually give us as a variance it will define the variance of this one so uh and once we have this notation we can simplify the cost function we had before like like here but now is the condition around ik plus one and it can simplify it into two terms the first term is about okay we it may be the uh the active inference we call extrinsic value the second one is the intrinsic value the first part is okay i want to move my robot location to that believe the source location which is denoted by the

normal estimation of your as a targeted location you want to make this error as close as possible so this is the task you want to perform something to basically move your robot close to where you believe it might be the second is how reliable disbelief and we quantify here by using the variance and initially remember those two relationship is derived from the very best the cost function there's no weight between the u_2 is naturally this is the optimal way if you minimize this naturally the optimal way to do it so that that this the cost functions consist of two terms here which is uh the task of the objective you want to perform and the uncertainty of your belief so you have that extrinsic and also interesting values and optimal to combine them together okay so now we can go to a little bit more detail about the equations and i will not go into too much but it basically is somehow like this is the best agency equation like as a next time the robot position depends on the your actions and also you have the dispersion model which is somehow like a ghosting model but it also depends on a number of parameters which is uh the wind speed wind direction what kind of chemicals you have they have some lifetime uh in the in the in the state in the air and then some other uh parameter associated with the location of the source but also we try to model the sensors and you should we should know that in the uh in the environment the chemical and also those viral agents their kind of changes are really low and many times it may be not able to detect anything but also you should also know they have lots of local turbulence which upset your sensor so that it means a lot of time you couldn't follow the gradient method try to say oh i found the gradient i found the maximum concentration no it didn't work this is maybe what it looks like after the true dispersion fields you can see the concentration change quite dramatically so that's a challenging so we also need model in the sense of behavior somehow like when the sensor you can have a reading so it's true reading plus some sensor noise whether many times you don't have sensor reading and so you just have purely have a background noise of your sensor so this is another point and once we do that and we have the uh try to try to see what are the unknown parameters about the environment of our source one estimate so basketball include the wind speed and the direct direction and and other things associated with the chemical components or i know them that parameters associated with the target i think it is the source location the release rate and then we can be using uh in our framework we're using basic uh inference uh to do to do it uh try to estimate those parameters so this is what the diagram looks like and so basically you have two parts one is about the reasoning another part is about the planning or control action in the rhythm every time it takes a new measurement and you based on the prior information and the and the model you have this way more and you try to update your kind of visual estimation

and then you build up a previous map about your local environment and then you fit this into your planning uh that here you try to estimate if you take any con action what's the influence about your belief and always how to make your uh agent close to the source so you do the planning here and is this somehow like this red box is somehow like that you tried using your virtual or a moment to do the reasoning and repeat this again and give you a dedication and then just keep doing this and so you can do some simulation study about this for example if we start from here this is source this is a kind of changing and then we can put the source in different locations and as a agency different location to see how they are performed and then try to understand the behavior performance and what we can see is that if we quantify the performance in two ways one way is we call the successful rate which is a and if we run the simulation 100 times or excellent 100 times uh what's a successful rate so for the new approach you can achieve about and for the uh some entropy based original hour method you can achieve about eighty percent chance found that if we're using uh classical model free control you always have about around eighty percent of chance to found this another uh the mayor is about how quickly your conversion move to the the true source location which is denuded by distance uh from the agent to the source in terms of the root mean square because you run it many times and what you can see uh the our approach is actually can decrease very convergent to the source very quickly and but others is open slowly converge into that and we also can do the experiment this is exciting bit you can put on the physical uh robots try to let it wrong and then you can do that using the software to implement your high level argument at a low level you also have a control loop try to command your robot try to follow drive it and follow your high level uh the division to see where you want to move that really will follow that we're using the sensors and also and some level control to do it so here's some uh explaining result you can see it's very turbulent and the conditions change quite quickly the robot starts from any points and there's a purple point giving your idea about is that believe where the source smile being represented by particle filters you can see with the time goes they're concentrating too uh closely and then to the um the location of the soil source and this is uh after that you can measure the area give you the idea about the concentration uh in this particular area and also we can do that in flight outside using your implement all the strategy on the uavs and the real test you can have the uh chemical sensors here this is a and all the other gps the cameras and all the sensors and you have a low level uh the ground control station are here and then you can do all the experiment outside they can this is a trial on the chemical plants they have a leakage of the gas and then what even happened and you can see

uh the test and we did it the industrial side and the uav taking off and they got the chemical sensors on that and they they don't lose and where are the sources they're based on that they give up some search strategy when i first started with we didn't do the intelligent research we just let the uav to fly around try to pick up the data and then we moved to intelligent research and try to demonstrate that so that pretend someone died because the the chemical leakage and then you need to send the uav to to identify where the uh the problem and then first respond can take a proper action to this so then we can map in the area this is not the intelligent such one but it justify wrongly to connect the data beautiful understanding you can say where are they are but then we move to a fully uh intelligent search and then is somehow like hands off we don't have anyone that's that is fully that you're able to decide where to find to connect the data and them and the child understand where they're so smiling could it be and this is another scenario i can see here and you have the car here maybe this is simulated environment you have vehicle involving the accident whether you have any uh petrol a liquid to that because this might be it means you have a risk of the explosion under them if you send the first responder to this area that were under the higher risk so if you could have an agent to be there try to search uh where have any uh potential and leakage of gas under the um where there might be then you found found them so that's uh and there's some idea about it and doing the this kind of research and and then we also and develop this extent this one to more broad applications a particular one we call the self optimization control that the idea is for any uh or mozart system you want to maintain it operate in best possible way which can change with the environment because the way the environment changes the best possible operation way for the changes however you want you are you have a consistent and by taking the same idea as we already talked about they're able to explore the environment influence on them they understand how to best operate themselves so somehow like here just change your environment and reward the function you just try to follow that and then try to do the best so there's many many applications i just talked about one this is a situation where we have you is about reliable energy you have a pv farm uh the energy and the aim is very simple you try to harvest as much energy as possible but the problem is the optimal operation is changing with the environment with the sunshine the temperature the the the sonar insulation has changed your operation optimal operation so what we developed is a strategy no matter what kind of weather condition you always can maintain the solar farm open in the best possible way it could be so the red one is ideal one green one is actually what we uh the blue one is what we did you try to follow the optimal this is optimal operation we always try to follow so that's

some examples not only for the autonomous search but we can use australia to solve much wider range of problems now we talk about that relationship with some others uh existing work one is sunday is a due country concept why we call dew control because the control action is not only changing your physical behavior of the u.s system but also changing your belief of the world of the environment so this is why you call it due control but it however is not complete new con concept in the control community we did this before but many on the dynamic system are estimate your own state how to understand some of your own parameters the idea now is try to extend from uh uav for example itself understand itself it's more to the environment because we think about it autonomous driving for the uavs we have all those information about ourselves what we really need is the enviro information about the environment answering about the environment another link is uh with the uh the active uh in uh inference uh the the community and this is a very uh interesting surprise funding from me because when i developed a strategy i don't have any idea about this community and is that somehow like you can start from completely different area different angle but however you found that you're landing on the similar kind of idea or area so this is i think one of the most exciting things in this research so so basically you can think about it is that uh that you control the idea is also about that your action will change uh your belief about the uh uh the environment and the you then connect the information this will again change your belief about that so there's the interaction between the action and the perception and how they're connected with each other so i will not explain too much especially in this country people maybe must understand what i'm talking about and another is about it it's linked with us yeah it's linked with uh the world by the way there are some other research in this area another is about the reinforcement learning and this is my view about the link within reinforcement learning so basically the reinforcement learning is trying to make an optimal dedication for a given dynamic system and subject a reward function and then it proven itself converging to uh found a solution for a beyond equation and then what a reinforcement learning though is how to solve this problem by the optimal value function and also try to find the uh option uh personally found that optimal uh policy so idea you try to learn the strategy by through the iteration but what we do is that from uh the work i'm doing is developed from something we call that modal project control we try to solve the same problem but however with uh truncated the infinite horizon problem into uh finally the horizon problem so every time we try to solve uh find the horizon optimization problem find the uh optimal solution but uh reinforce them early and try to do using the iteration uh like uh uh learning try to do that so basically

and and the whole idea about link between them is actually explaining this paper and i just the publisher if anyone in the community interested in this and again please have a look this is my view about a link between those two and also it's a link with active interference so and because the approach are different i just given the conclusion for the reinforcement learning they have three major problems i think about it one is they need a lot of data to help them to learning the optimal strategy opportunity function and the second is once if you learn from the simulation environment when the environment changes and oh there is a mismatch between your simulation environment the real environment the optimal strategy learned in a simulation environment will not work very well in real life and in particular in this case if i learn the try to search the sourcing based on this environment okay the wind comes to this direction speed i learned how to do it but if you come to real emergent situation the wing is actually blue to this direction the strategy you learn maybe doesn't work at all because it's not optimal another problem is actually that being frozen like black box is very difficult to prove the stability or safety this kind of thing but we don't have in the control we have a rich body of tools we are able to prove the stability safety and other things i think the first tool is able to solve by active inference as well like a new reducer the number of data required and also that have to deal with the unknown environment but however that controls also have another uh office which is about safety about this so this is a quick try to think about how the link between the approach to talk about here and the reinforcement learning and and then we can discuss more itself for this particular talk then the the the kind of work whatever we are doing we try to move this from single step to multiple stack looking ahead and the final horizon but this will have a problem about a computational load because now we have to deal with a much higher capability how to reduce the combinational load is quite difficult and and last time when i talked with a car freeze it gave me some ideas about the team how to do it we are trying to learn from your community about how to reduce computation alone another thing is what we are doing is we try to prove that a sense of rigorous properties for the approach somehow like to try to prove they can converge into the two sources and there if you do anything there your belief can converge into true as a the external environment under them and and also and you are able to prove the safety of that which is particularly important for our area when we deal with cars and aircrafts we have to prove it is safe to do it so we are working on in those areas yeah so um yeah conclusion is uh we develop some new framework in our way and we try to make system can operating on the environment by uh somehow like a trade-off between the exploitation exploration and try to understand the future action is influence on our belief so

this is uh can deal with the coupling between the action and the belief and our approach is not somehow like many other approaches they are saying about uh if you want to trade off your artificial stitches uh uh interesting or extrinsic values together add some weight and it's actually this is naturally it happened this is a it is optimal in some way and the particular center for us i think about it is we have a particular aware of the community of active experience we can learn a lot from there because they community here people working in this area develop many good ideas and the question is how to send it to that how to promote more collaboration between those two communities i would think it's one of the very interesting things to do so yeah that's all of my talk i think i very much come to the end of my time yeah it's 29 minutes now yeah so that's good okay yeah thanks yeah so i will join the roundtable yeah thank you yes okay thank you i will leave it first and we do join okay thank you all right well welcome back everyone so as it turns out we may or may not have any people join we will have just one join right now and then we'll see who else joins and so welcome back when what was just saying that anyone who's in the live chat please feel free write any questions in the chat and we'll be looking at them when hua not sure who else will be joining us but i think actually this will be a a great opportunity just for as short as long as you'd like to talk a little bit about a little more about what you presented on and also connect to some threads that were happening earlier in the presentations how does that sound and we'll see who else joins sound good i actually don't hear you anymore okay one second we'll figure this out yes alright so we can hear you now um yeah okay sorry all right welcome so we'll talk about some of the previous presentations not sure which ones you've seen and otherwise i'd like to pick up on a few threads in your presentation and then we'll take any questions that people are asking live how does that sound yeah um awesome so one point that you highlighted was this notion of dual control and how it brought you to some of the similar places that active inference has arrived at so i'm curious what you think led to that what led to those vectors intersecting in a dual control and what is at that nexus or why does it exist that way and why is it coming clearer to the forefront at this time yeah thanks for that that's a very interesting question i know also i i also think about it myself from time to time i think that there is a certainly those two areas i feel original was quite far away uh we have different strategies different way to think about the the world typically control we normally we are interested in uh as i said the due control concept uh is not entirely new and uh the people in the in the control community already realized that many years ago somehow like if you take any action the action we are not a changing behavior of a dynamic system but also maybe you could have some influence on

the variables you are interested in in a traditional control this could be for a physical system there are some parameters yourself like a mass or a damping or other uh things that you don't know you by doing that will help you understand what you really are just like when you're driving your car you have some river you know what i mean or inertia my car have by doing some certain action so the people realized that uh before but the difference is now because we control the move from uh typical maybe can talk about low level automation now move to high level we call that intelligence or automation uh autonomy okay now you move to the more higher so then that means that you cheated that dynamics is like more like an agent in our way in our languages computer science or neuroscience is an agent so now we're more interested in not only about ourselves more interesting about the environment surrounding us the outside so my the work i think is now progress like what i did is move from you're concerned about your agent yourself more about what are the surrounding you how to explore the environment how to understand your behavior in the environment so by doing this i feel we bring the typical engineering heart sorry the control in feed into this kind of natural water in this kind of biological kind of sense because now you're more interested in how to deal with coexisting with the environment how do we know so this is why i view on this and but on the other side this is also what i i reading on the on the active inference but suddenly this community have more say i feel traditionally before the due control people already have some idea about a treat human brain as a basing machine so now he said okay i have prior information about the environment i have a new data coming helping you understand what is happening around me about the environment but i think what active inference for me i think about it suddenly the other wrong side of how we define the reward function and in particular talk about it using free energy to quantify that this is really wonderful to the world but also i think about it is key things here is is about you now think about it i action the action is also linked with my uh understanding of the environment about perception so the carbon so using the action to explore uh the environment gives you better understanding so that it means you move from just that to the human as a passive way to receive the information from the environment nowadays that i can actively do something helping me to better on the same one so i feel this may this is my understanding that the key feature of active inference so because this is the reason you can see the naturally will link with the control we talk about it because the control normally talk about what is consequence of your action how to take the best action so this is why i feel those two areas they from quite a different area gradually because the need for the research and also something like a trend in the field they are now moved together in

this particular area they come conversion to each other so this is what i'm thinking about awesome if i can just like pull out a few threads that i i thought were really insightful there you you cited a paper from the 70s about this dual controlled notion the idea that we had to have not just like a single imperative or have an imperative that contained epistemic and pragmatic components and all that that entails like the unknown consequences of action so what we would have is like expected free energy as opposed to the variational free energy and then you you described how there was a movement from lower level automation control systems the classical thermostat and other kind of like um single variable yep single fixed point implementations as the implementation complexity approaches the multi-joint and way way way beyond how that multi-joint is going to be in a social context or a changing environment system that's being designed comes to convergently require the same imperatives of a nervous system which is to say like the real-time integration of sparse sensory data and also that is what authorizes the ecological stance and the embedded cognitive perspective which ties in with some other recent threads in cognitive science and neuroscience like the pragmatic turn and so it really is interesting how like from the technical capacity side the questions that were converged upon in terms of memory and forgetting and all these things are like the challenges that natural systems have been involved in solving for a long time yeah and that's absolutely right that's a very very interesting part of their technology develop but also there is a maybe demand from the society and because of lots of things like of the traditional mechanical thermal kind of a simple control already while existing they help us to boost our productivity for many years but now we reach to the stage we want to further increase our productivity increase our efficiency you have to develop a highly automatic system and which is able to deal with some unexpected events uncertainty and so in order to have this kind of uh like capability they also require a high level of intelligence like our human or animal so this is actually another reason the technology of the move is because the society and economic the demand for this kind of thing to happen okay i have another question about your uh scenarios that you explored how is explore exploit balanced automatically how can such a large statement be made and how is it balanced when risk is in play like risk of going into a dangerous area so within the search task how is explore exploit uh mediated and then how does that become more complex or how is it integrated in the model when bodily injury is on the table yeah thanks for this yeah that's very interesting question and uh also we try to and think about this along ourselves for a while and the first i would like to say for just using the autonomous such this particular examples to illustrate the key ideas the the fundamental

objective for this particular task is very simple or clear somehow like based on my understanding i try to move our agent try to move close real life to resources okay so under them they will formulate this as a reward function or cost function somehow like you want to move your next times agent the location to the source location as much as possible but however because you don't know that so there'll be conditional on all the information it connects so far so this is a the reward function your origin defined but from there we derived this particular function if formulating this kind of way and then and also try to introduce the the note about it the action will affect our belief and then we'll naturally you found the cost function or the order function becomes two terms won't consider two terms long term is about the exploitation another term is about uh exploration so there's a natural happening there is no like you need to introduce some terms or they just naturally come together in this kind of way so this is why we think about is maybe that is the best way to try to do that but also you you mentioned a very good question about the risk for example if they naturally you're not only something like a target they want to follow but also during the process you need to aware of kind of risk there so normally you have to deal with in our framework in two ways one way is we have to add some constraints in our action generation it means for example one example is in our search if there is a obstacle in some way if you command your vehicles or aircraft move to that direction suddenly this risk because the client is going to collide with the obstacles we added the constraints in the search domain you said in this area you couldn't go this is one way to do that but also you can add some we call it called softer constraints that means in the cost function you add some penalty in your cost function said okay there is some area might have some risk you try to avoid if possible and that when you generate or optimize your uh your action you try to take this into account all right awesome i want to connect that to active inference and then ask a great question from dave so you mentioned various constraints could be provided to prevent like situations of of risk or hazard to the entity and so that is uh what may work in practice and so that's why it's been so interesting in the presentations to see that most of them have featured at the very least um a laboratory robot context and that kind of makes the full stack make makes uh increases our confidence that the full stack kind of touches like there's like a tesla coil like there is a path through some implementation and then the question is how deep in the specifics do you have to go versus how um far up in the generalities and then there might be some situations where the constraint is applied in a very situational way but also as you laid out dual control and reinforcement learning you said reinforcement learning is not really amenable to being formalized

analytically in practice whereas um one active inference strategy that we've seen to balance like a task performance with survival is you say i want to reduce my surprise about the gas cloud and i want to reduce surprise and have a high battery percentage so then it will just like balancing explore exploit within a drive like to detect the gas or to stay high battery it also can have a nested model that's balancing those drives and so there's also an analytically simple and first principles way bring homeostasis and risk avoidance into very generalized framings of the active inference framework and then again in the specifics it might be useful to do different model uh variations closer to the edge but then it's very interesting to think about what it looks like when also there's something in the center that has a simplicity to it yeah yeah yeah thanks for that that's a very useful discussion are different ways we can deal with the balance between the uh you want to do what you want to do and also try to avoid any risk or beautifuls and then try to balance try to deal with particularly i think the problem here my feeling is called uncertainty because on sunday will have affect all those things because if you think about risk how relia how possible reliability this risk might be so there is a lot of things like this my uh this will come back to my uh my fellowship my fellowship was funded by epsrc for five years i concentrated on this particular area the goal of my fellowship as we already somehow indicated we tried to increase the level of autonomy by if there's more intelligence arguing into the control and and on other field but one of the the idea for me is i we could want to develop something we call that goal oriented control system it means we want to promote because high level of intelligence like us people or animals they are more have a goal-oriented behavior you rather than just people you said oh i give you the goal you try to figure out how to do it if you have a next intelligent one he said oh i need to give you instruction every step about how to do things properly so the first to promote the goal in the behavior i said okay for this particular what is the task what do we want to achieve what a requirement so then second things in the key integrating my framework is about the constraints the why the constraint is so important because you try to avoid any like before any risk of things or for example if you want to develop autonomous driving cars you have to follow the rule of the traffic at the road and they have to follow that and they have followed the traffic light and other things and they and you don't want to collide with any other vehicles or uh or hit any potential so those are the constraints you have safety concerns in this case but you also have a lot of physical constraints because if they have a maximum power maximum like temperature or pressure your system could have otherwise you are going to destroy yourself so constraints play a very key role here but the sunday the question as you said four

different scenarios how to abstract formula this constraints is a question but the whole idea is you have to for any decision or action you made you have to respect those constraints before you try to think about oh wow won't achieve any uh meaningful objective the another key part of this one is uncertainty the onsen they came from many many ways could be the environmental changes or onsen we already talked about it but also could be answered about your information because you have a sensor sensor could have error or the sensor range is not enough to pick up all the information you need and there is a yeah so there are so and also you answer their body your belief of the world for example we talk about the risk but how reliable this risk it'll be so that i will think about it and this is my belief in the whole framework for uh you control the key thing is about how to quantify the answer about your belief of the surrounding this is driving you are you rewarded expert exploration so those are the key things in my view and basically if what our idealization is think about you talk about in the active inference we are able to formulate that in electrical way and then we are formulated in electrical way we are able to using some tools or theory people developing the last 30 50 years in particular in the control community and other community we're able to formally prove if we do it in this way we can make sure you are able to satisfy certain safety requirement it will not hit the some obstacles you'll not put yourself in the danger in the for example like you eat in the natural world either by other animals for example so so you this is the dream we have some progress in this area we're already able to do some a very simple system we are able to prove stability or safety of the uh that you control but however is and we want to do is try to extend a much wider world and therefore the free energy principle is much more difficult because the energy the functioning level is quite complicated how to understand that is another level of difficult but however in principle we are able to work together to push this thank you there was really a lot in there i'd like to make one remark on and then bridge to dave's question in the chat so you highlighted uncertainty and that's of course a core aspect of active inference with bounding surprise using free energy and then you mentioned goals and constraints and that really seemed to me like a a common point with cybernetics and goal orientation constraints general systems theory cybernetics branch but also engineering and entrepreneurship and innovation as your colleague stephen fox has worked on and those perspectives on goals and constraints of potentially nested systems of organization like projects within an organization or a firm within a market or a cyber physical system nested within yeah to have interfaces that can even just be described in an information partition way and then you brought it back to safety and to be able to have certain like probabilistic or

formal guarantees on those kinds of arbitrarily composed systems is very exciting direction
um any remarks on that or i'll ask the question from dave i just have a quick remark i think it
really for me i'm more on the technology side but however is absolute this kind of goal in the
behavior you can using for the social organization for the biological or many many other
things because basically when you survive you have a goal this goal you try to survive on the
food and then the society you want a new organization the goal is to try to find more profit
constrained by the kind of marketing environment and many other things yeah yeah i
absolutely agree with you i'm just gonna make one more comment to refer back to an earlier
talk not sure if you saw this one but this was tim verbellien's talk and he was talking about
video games and about how just with a only a epistemic drive some of these video games were
able to be played very well like with pong and with mario and then it made me think about
how in a lot of video games staying alive is an imperative whether you're just tapping
something or a maze or something that's growing or shrinking it's like staying alive bridges the
gap between search and exploit and all these different behavioral modes it's like if you're not
staying alive if there's no like idle process for your cpu then it's over so then yeah there's no
point in future epochs so that's really interesting so let me go to dave's question so dave uh
wrote professor chen i didn't notice discussion in your presentation of of affect or emotion or
drive what do you think about adding such mechanisms to the dual control model could this
amount to explicating internally generated motivations or would such a trick be a mere
disguise for inserting arbitrary experience experimenter specified rewards that's that's very
interesting question and when am i reading the uh literature in active inference and i'm fully
aware of that the focus of our research is quite a layer and we focus on do something uh
specific or useful and in this case for example in the case uh the autonomous search is just try
to find the location resources and in the in the pv farm for generated renewable energy the
object is quite simple said okay we want to harvest as much energy as possible given the
condition so in that one means the focus of our one is a very specific but however when we
come to the biological and also human and others they have a lot of things like the the emotion
and many other things we didn't consider in to in our current work and i i can see this is the
gap between the due control and the active inference is actually you can deal with talk about a
more general uh skills like empower or confidence or some other more things emotional things
and and this is actually the direction of the maybe the future of the and our work should move
try to learn from this community over that but in principle and as i said if we are able to
formulate those kind of emotions or many other things as kind of a reward function for

example you you encourage people to be happy to but you need to have some way to quantify that if you couldn't uh capture that maybe you will not promote the behavior uh try to get make you happier so that is the key things for us is how to uh formulate uh interesting uh reward function and and this is also what you can learn from active inference in principle and we didn't have a restriction on what time what type of reward function should it be what type of constraints should it be the framework is quite in some ways quite a general but however for more complicated systems still did a lot of work yeah that's very interesting if i could uh build on that so i was imagining the setting of firefighters and there's some different chemicals that are being sensed and about the way that the extended cognition paradigm and seeing the cyber physical team as being just qualitatively whatever tools they're using even if it's just the walkie-talkie and their bodies it's still like an extended cyber physical team and then as we move forward into futures that may in different areas like include all kinds of tools that we do and don't know about so then that's what's interesting about a framework that can start and pick up in the um like qualitative zone and then also take it all the way to the application and then as to value alignment i thought that what you just added had um some seeds of new ways to think about the human robot or just more generally the multi-entity alignment question and that is their preference vectors or some of their preference vectors could be about the same thing and in the same direction like the semantics of what the firefighter on the survey says i want i want to like balance x and y and z in this ratio and then there can be an entity whether that's another person or some kind of explicitly structured cyber physical entity that also has that same like preference distribution and then yeah that is like a formal way to find that probabilistic alignment because the question has to be addressed one way or the other and how will those distributions align accidentally or using some other simple heuristic yeah that's absolute is actually when we talk with our end user and also firefighter and also the minister of defense they're interesting therefore the terrorist attack or some other kind of things you know in london in the tube station there's some scenario like this and here they'll talk very rarely just send the robot out and let it do off do all the things he's actually always have an interaction between the first responder and the robot really want to pass the information about oh where i thought it might be and then the the the operator first to provide some extra information and they'll always work together as a team and also there's some preferences for example in some areas based on the first responders experience they knew where they are more likely could it be that when they see a picture or see the scene so there is always the interaction with that yeah absolutely this is actually our future step is somehow like to promote this kind of

interaction between the human that in in some interference and many other emotional other urgent things because some area if you close to maybe exit or somebody you maybe have a priority to search or find whether there's some probe there so okay a few remarks on that so it made me think about how the response network like of an emergency could be seen as embodying a prior like where i am in california maybe firefights are a certain likelihood different seasons so then um a phone call to emergency services might have a different likelihood of something being the case in one region versus another or it could be kind of used in like a bayesian filtering way where are we getting 15 reports of the same thing and then where are we like we're already working on that situation and there's something else that needs attention to be brought to it so we don't want to like have a sampling bias and then that question of how training data in a static learning context results in basically like biased implementations in the real world was was a theme of several of the talks um a question i wanted to ask was about the turbulence of the flow it brings a chaotic multi-peak landscape because what i saw in the cloud was there's like it's like island chain it's not just one ridge and so um how does a smooth gaussian variational approximation make sense of something that otherwise we might think we would use like complex simulations to resolve i think you're a really expert that's a really good question and we struggle with this for quite a while and you can think about that you have a lot of local turbulence is there and also uh com coupled with any case if you have a chemical biology dispersion the concentration of the air is actually really low and and also for our uav the small drones or the uh the robot the sensors they carry is not very advanced you can't carry some level uh grade very comprehensive equivalent is actually we use a very simple uh gas sensor so that means you have a high level of that mis detection or and and also this what makes it even worse is because you talk about it that the turbulence you can make and when we operate on the uav the propeller didn't help us it itself it generates lots of local fluid it's upset our sensor so it's hard now it's somewhat appreciated lots of hard work getting into this area but but try to answer your question is and when we uh reasoning we could use a more complicated model but however the complicated model sometimes didn't give us in this case didn't give us too much benefit for to reading one reason is uh the complicated model is always have a um because the environment have a high level of uncertainty and even you have a complicated model if it's accuracy to represent a real environment then that's a complete model give you benefit you feel that come the is environment is so much uncertain there maybe the simple good enough model is able to do the job there's a first question but also by using a simple model it can signify driving you a

computational lower down the simple gaussian mode it is very simple but however you are able to driving your computational lower down because compared with a safety model computational fluid dynamics mode or more complicated model uh second is because there's a particular scenario the uh the sensor is not high grade so a lot of onsen they caused by model maybe uh just the uh display within the sensor noise because that noise is quite high even you try to push it more and more and more accurately but it's just maybe one percent since the sensor lawyers give you five percent so it didn't give you too much benefit this is what we found in this situation so i'd like to maybe connect that gas dissipating setting to one of the earlier talks and again not sure um if you saw it but just to kind of recapitulate the point so this was from tim schneider's talk and tim spoke about goal-driven active exploration so i'm sure that there would be a lot of resonance there with the dual model and one of the questions that he really highlighted was he asked can the familiar have intrinsic value because he described this uh problem called the detachment problem where like a region of high posterior likelihood a good region to be is getting walled off by like a well-explored region and so that dissuades further epistemically driven actions into that kind of ring it's almost like we've already been there but then it's like the you know the the new finding was just one more paper down that down that bridge um yeah and then we talk a lot in active inference about morphological computation and body computation and i think in the setting of the gas dissipating there are several ways like we can be really specific when we're talking about applied embodied computation so one was what you just mentioned with the uavs and the fans and how that was causing a distortion it wasn't just like a free-floating sensor and that's like our own body models and so people talk about how the body is a model and it has a model of itself in space and that's how we can do all these actions um and then the second example of like embodied and extended cognition was the way that the gas was dissipating so it's almost like the forgetting in the model was happening automatically because you couldn't just build up a heat map of where of you know the integral of gas flowing through a region of time you have to have something that is dynamic but you're local searching a space and so your estimate in other areas that are distal will become increasingly uncertain so yeah what does that make you think about in terms of the work that you're doing yeah all those things are very interesting there are many directions for that and we we think about it the work we're doing here is just try to illustrate the very best principle about how to take the advantage of the um the action you take to give you the new information and then from there to help you so this is basically as we talk about bear the same spirit as a active inference so this is the fundamental things and

in terms of scenario and the complexity there are many many layers so we can add the things on and also one thing you mentioned about the uh for example the influence of the uh the agent or the robot is dynamics but also it could be for example we consider a source now is in the in the stationary is fixed they are just released the gas but in the real life maybe there is a mobile and uh things that people some terrorists for example put something on the on the truck or on something the driving around or some other uh things could happen and there are also people talk about uh intermittent release sometimes release and stop and then release again so there are many uh much more complicated scenarios here and could be very interesting um coupled with some other things things people talk about for example you could have more than one uh sources in this release and it's just not one single uh sources could be have more than one and also people talk about if you have a larger area and you have a number of agents try to work together now we talk about the collaboration between the different agencies how to work together more efficiently to to search the area so there are many many things here and the one particular thing that i feel is directly can get from your the help from your communities now we only look at a one step ahead but in real life we already have some similar if we look five step ahead it actually give us a much better result because you look further after the influence of those changes so so there are but however the the the the downside is uh we'll already talk about it is a about a computational load so now you have a like a tree search because each step you have a number of directions you want to move your agents and you very many uh multiple steps and the accumulation could be quite nasty and we now have a researcher trying to think about using a multi-colored research or similar kind of and the car is also provide for instance provide some idea about it and you work in the area uh how can embedded into uh our work and try to take the advantage of this so there are lots of interesting things here awesome i i'd like to um i think just to create a few of these points of contact pick up where you said that you're planning one step ahead um which is analogous to the variational free energy it's like the instantaneous best action and then the expected free energy takes into the future and i wanted to mention a few different ways that temporal planning is accomplished and some were mentioned in the earlier presentations also it's interesting to note that however many years ago i don't know the exact number but somewhere in the 10 years ago range active inference was kind of like an instantaneous perception cognition action theory and several elaborations have specifically enabled it to account increasingly distal planning so one example is like in the continuous time having a taylor series approximation of the generalized coordinates of higher and higher approximation depth

is one approach another approach that can happen in discrete time is having just a broader time horizon for policies as well as with different uh tree pruning approaches another approach that can work with discrete time or with hybrid models with discrete and continuous time is hierarchical modeling yeah and so it's very interesting to wonder whether for planning a hundred steps out to go to think really deep into how the uav will do something far away how does it chunk that and how does that chunking into is it a hundred steps or one ten steps to ten two steps of fifty and five of ten inside the way that it starts to chunk and understands are the ways in which the computational burden is reduced and also those ways that start to resemble the ways in which biological cognitive entities also make sense of their environments yeah i think that's absolutely right that's a very good way to move forward we're also doing some thinking this way but not necessarily coming from the same direction because we think about it for example when you make a decision on on on the for example using the uh autumn search problem you somehow like you try to design your your way point where you want to move to that and then suddenly you have a question about how far the step size this exactly can change and even the larger area maybe you want to move a little bit longer distance so somehow this is a you can regard this as a hierarchical uh strategy and then we actually uh give the commander to low-level uav control or autopilot they try to follow those kind of things so there's a lot of things that we need to think about that somehow like is trade-off between the performance computation load and the horizon you won't be looking ahead in theory if you're longer it gives you better performance but however how far you are able to looking at so there is a trade-off between those terms maybe four different applications could have different factors but the whole idea i think that's quite interesting is exactly worth to explore cool um have also kind of to to pick up on some similarities and differences with the some of them utilized neural networks as modules in their training other used variational bayesian methods which can be fit as an optimization problem and also we saw sampling based approaches so i guess just pretty broadly how do you see these different ways of robotics and edge computing how do you see them working in different situations or together sort of pre-trained or updatable models that are large or variational or sampling based approaches yeah that's that's a very interesting topic and i was quite abroad and there lot of research nowadays about using uh pre-training the model and particularly in the context of reinforcement learning for example if you learn things and then try to deploy them in the real life particularly for the other robotics as well they have lots of research in the area and uh for me and as i didn't have any time to explain uh articulate those ideas fully but i think about it is lots of problems whether it's

engineering system robotics or the biological or the the human is is however the the the city could be described like in how to uh like make a optimal division at any time the optimal division could in terms of try to find a food or try to survive or try to do something useful no matter so that overall those all this problem in my view can be summarized as a optimal decision making problem for this one even when you have all the environment information unknown and also all the behavior of your agents unknown is not easy but when you try to deal with something like what we talk about is uh uncertainty in the environment on the environment it will become much more challenging so and uh there may be there's no time being no uh like a single solution for this this is why you talk about many different approaches they try to deal with and the problem for me is um uh there are two major uh approach i'm thinking about it one is i call the uh iteration process the iteration process in particular like in the reinforcement learning what it means you have some initial strategy you try to take the action from the environment and reward the function or set the change you learn from that gradually somehow like you make your uh policy uh or or very function close to true truly the optimal one option policy so you're always doing that in uh in the in the iteration learning learning learning so this is a approach and uh and but because this approach you need a lot of data this is why now people tend to train in the data i train is australia beforehand and let them deploy that maybe online you can adapt to a little bit to that but you if you can using offline training using a lots of data to take the advantage but there another approach like maybe more like in the uh in active interference or more like in the due control i call it as a purely like a online optimization approach the online optimization process just like said okay giving all the scenarios giving all the information i have with the about the environment giving all my what is my objective i try to work out what is the best strategy i should have given all the information i have and on the body environment about my reward function i just try to work what are the best strategies so then this problem can formulate as an online optimization program you try to say okay give all the information connect so far i give all the understanding you charge it to that and uh for this approach uh you need to have a larger computational load normally as you talk about aging computing and you this is why we struggle we talk about a one step ahead because once never had these means is the optimization problem easier to solve if you look at a multiple state the optimization problem could have getting much more complicated and aging computing maybe you're not good enough try to do it to do it but in principle i can i regard itself as a as a two broader approach to suggest that and in the inactive inference this is precisely you tried using free energy or expected free energy as a reward

function you had optimize your action so to give you the best possible one so so this is why i feel is this kind of a similar kind of approach as we talk about it so i i was also while you spoke to the dual control systems thinking about how it might have formal differences from free energy principle and active inference as we know it today or again like we discussed earlier how it might have reached some of the same points from two different angles and it made me think about the particular partition and the markov blanket or the first in blanket so do you have any thoughts on this or i can provide any more context but first i just wanted to ask you if the concept of blankets was relevant or there was some analog in the control literature from your perspective yes we we maybe we can say we always do that in our control community so we always do that in the sense traditional m control look at is our own dynamics itself and the sensor try to pick up the information then the action will take so this is a controlled way the sensor and your own thing and then take the x but however previously control we very much like ignore the the interaction with the uh too much of body environment we talk about environment in the control typically we call that disturbance so the aim of our control system when we are existing is try to act against any uh disturbance because we try to uh to do something ourselves or keep it for example if you think about want to keep your room at temperature at a third degree if some open door is a little if the external temperature changes all those our control is somehow like i don't care about anything outside i care about it sensors take information i do something under them i take the action so this is always regarded the water is information from the sensor and we take the action which will have some influence outside then we take the information from the sensor so whenever i think that michael blanken i found that there's a natural and maybe some somebody close somebody with some thinking we we have i feel this is quite interesting the way to looking at the world and because of you that you in this way you don't care about it in someone open the door don't open the door you don't know because you just say from the sensor what's the temperature changing and we we take this ignore anything outside that i'm glad you added that because i also had wondered whether the control loop was like a pretty active partition i mean input output systems systems with interfaces holographic systems more and more equivalences have been found but it's kind of like the sparsest representation of interaction between two agents or with an agent in a niche is like well you know you need the road going one way and you need the road going the other way otherwise it's not like you haven't closed the loop so you have those four pieces like the two entities and the edges and then it is interesting with the way that what comes from the environment is described is it a perturbation or disturbance something

that should be controlled as the case with allostasis and like biological processes that are anticipatory and then going all the way into like novelty which is like the ultimate anticipation it's just anticipation of the different and then it's like again paralleling the development of industrial control systems going from stabilizing vibration yeah into needing to be proactive there's also again a natural coming as the attention in these fields turned finding formal cognitive models yeah yeah absolutely i think that the the principle of the market uh branch i think i quite found quite interesting relevant um but they are the the big difference from traditional and uranic cultural thinking to what are thinking in your area or a current our thinking is about how to represent the environment in your brain so how you try to say this is that make the all the whole things different as i said previously maybe you don't care too much everything like today the disturbance or are just interesting that your heating system another and now is actually no i want to have some way to if you want to anticipate something happen it may have a better implementation about the environment and also you want to align your belief of or inside state more close to external state so that is where i think all the city things comes and also where you are able to make some high-level division make a where the intelligence come from because you have better implementation about that you can figure out a better way how to do it and then how to get the benefit from the environment one example that made me think of and also connecting back to the embodied intelligence and the morphological computing i remember seeing a a robot or a little vehicle and it was able to move over a surface that looked kind of like a stair on its side like it was just very jagged um but the wheel that this machine had was shaped like a square that was the right radius so basically it was able to roll perfectly smoothly across that one frequency of stair but a different frequency of stair it would have just been a total total bumpy ride um yeah so if the wheel would have been small then there would have been a debate around well should we represent chunks of stairs and up and down hill and all that fine scale but then when the morphology is fit to the niche in a way that's natural or in a way that is off sourcing some of the computation to the physicality then that entity only needs to consider like a linear movement as if it was like in a simpler space yeah and so then it's almost interesting to ask um like in some of the presentations they they used um very standardized robotics platforms like the quadcopter and the turtlebot and other standard to the extent i understand standardized robotic hardware which is awesome because it increases accessibility and it demonstrates it in a clear way like tomorrow we'll hear from legos implementations and so but then that element of embodied computation starts to show how we could almost work in a different way and ask what kinds

of systems could implement certain functions maybe there are shapes of robots or other objects that we we just haven't seen the shape yet they don't have to have two legs they don't need to be five feet tall they don't need to look like a trash can like that's some of the morphologies but then with the air and the water and the ground there's so many bodies just of insects so yeah it's it's going to be really amazing to see how this is used proactively to design morphologies and behaviors that just do things we haven't seen yeah yeah yeah absolutely right i think your abs are right yeah and in this kind of uh i think maybe people call uh the the fitness thrive somehow like this and that is actually somehow like we always maybe you can using free energy or some other notion to describe because you you're living the environment you want to get the best out of that and we talk about now is a strategy how to do it or how to but however in the natural world because you're physically uh happier you your things are gradually involved with you and if we are in our design we can choose those physical body as always as options as something we are able to change and maybe gradually can combine those two things together it's not only your brain to make the best strategy but also you are maybe called the actuators or your physical bodies other things you are gradually will change with that so this is about how you i feel how to set up the problem but what i feel much more interesting is uh in the active ingredients when you're using free energy and then you can try to capture much broader kind of behavior and as i said in the control area we now maybe more focus on the specific task specific mission this is what we do physically we want to do it but however a lot of things like uh soft skills and like more uh uh broadly about like our confidence or a capability we don't talk about it but human when we do the things we learn learn build up the confidence or embody our skills and then we do more things now this is a and i think this is a things happening there and active embarrassing able to explain that they able to have a true framework to do this we are maybe at a more on the physical alignment so i can see things that gradually can move together and you may can make robots for example more capable to do something if they have the ability to change their borders or change their activation change other things or even maybe they have come improve their sensing because there's something greater they learn and they found more important than other sensing particular environment they will try to more develop that kind of particular sensing capability so i can see that naturally and but the question is about how to set up the problem we allowed it to make this one fragment that kind of that reminded me of from my own area of insect behavioral modeling is there are many models of task performance like digging or foraging however there are fewer models that are describing task transitions even though those are

really important for the colony and then the models of task transitions tend to be more generalized dynamical simulations and less getting into the kind of agent-based modeling perspective and so it's like as we're seeing robotics with the physicality and the technology pushing the frontier that kind of task switching becomes important higher and higher orders and then again that brings us into the bio-inspired design conversation because task switching is so essential and the idea that like the human is this unfolding of memory and anticipation and all of these different features it's hard functionally or neuroanatomically to separate out the human brain works and how animal nervous systems work and so i think that'll be another interesting area with this tension between explainability potentially even austerity of the models in that they'll have parameters but the parameters in the way they interact even for a few parameter model might be difficult to understand because it's not like going to be all the information is in those generative model parameters it's going to be those model parameters and the dynamics of how gas is in the world that are sort of relied upon outside of the focus like the the background context for the model so the model itself won't be fully understandable or explainable because it's going to rely on this context just like no sentence can stand alone so there's just so many interesting areas and and um to see that the directions that different fields can meet at and then start to structure a productive relationship that's that's absolutely right and uh i think i also absolutely agree with you it's about the the importance of models they in many ways they are very very important for us we quite we i knew that in the machine learning area they have a model-based approach and a model-free approach i'm more biased to model-based approach because um you can using um like a model-free approach they are quite powerful in some ways but however maybe they suffer some issues you already uh highlight how to explain this uh to why you make this kind of deletion why take this kind of action and more explainable and and furthermore the best one because a lot of things we are doing and particularly in our area there are engineering systems they are have a first principle or other models already they are for many years but however is now is the question is about how to make a use of that and so i prefer modal uh based approach in many ways along for the explanation and try to you can either understand the action but also for the uh i call that um somewhere like maybe people yeah use the word sampling efficiency somehow like you can use in less data because you have a model based approach and in the in our engineering award that data means money i knew a lot of people say okay the google facebook can harvest billions of data online freelance but however in the engineering if you want to do something you need to experiment to test you if you're into physical wrong something to get a useful uh

data come out from that so that isn't the money so when we're using the model based approach is much more efficient in terms of learning understanding what's happening around the environment and so this is why economically i feel is quite important for this not just for the yeah so for for the safety or how to make it more explainable but there are some other reasons well very interesting is there any other remarks you'd like to add or questions you'd like to raise uh not really i think that's really a very good accommodation i i'm very pleased about this community as i said and it's a surprise for me and people share the same idea so what i'm interesting is and they is getting more involved in the primary community and also i would like to open the door if anyone working in this community they want to talk with us want to work together and develop some new ideas as we already discussed and they're more than welcome and you could have my contact under them try to talk with me and so basically i think we share a lot of fundamental ideas and we and also there are some different tools uh different slides different concepts and approach if we are able to work it together i will think about we make them not only as the tools try to interpret how the natural water and the human or animal the behavior like how but also make it as a useful tool to drive us like this theme to design more capable robotics and do something for the society amazing great okay close to the first session so you can depart wow what a great discussion big appreciation to professor chen for joining for that that concludes the first interval of the second applied active inference symposium on july 31st or at least it is now where i am in about eight hours we're going to have the second interval it will feature presentations by bruno lara matt brown adam saffron and jf claudia it should be a great set of presentations followed by a round table discussion featuring several of those presenters as well as carl firsten so hope everybody has a good break in the eight hours before the second interval and prepare some thoughts and questions and writes them in the live chat or emails us really appreciate you spending the time and attention listening to this active inference institute symposium and hope that you'll stay involved participate and keep on acting and inferring so goodbye everyone and see you in the second interval you